

# Geometric Invariance Priors in Vision Transformers: Principles and Costs

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**Abstract**—In vision Transformers, geometric invariance priors expose a long-standing tension: they are expected to improve efficiency and robustness, yet empirically they conflict with discriminative and scalable learning. A clear theoretical characterization of this tension is still lacking. To our knowledge, this paper gives the first systematic study to characterize the principled form and structural cost of geometric invariance priors in vision Transformers, offering the geometric interpretation and positioning of MetaFormer-like designs.

- (1) **Principles.** We use *localized Euclidean-similarity transformations* as a visually natural model of geometric compatibility: the same local visual structure may appear at different locations, orientations, and scales, while a generic learning rule should not depend on these local coordinate choices. We prove that, in a generic and scalable learning setting where feature channels carry *no* prescribed geometric structure, requiring localized-similarity compatibility forces any regular local operator to collapse to *shared pointwise channel mixing*. In the linear case, this result reinterprets the ubiquitous  $1 \times 1$  convolution in vision networks: beyond its engineering role in channel projection and computational efficiency, it can also be viewed as the principled form of geometry-compatible generic learning.
- (2) **Costs.** This pointwise collapse reveals the structural cost of invariance priors: geometry-compatible generic learning *cannot* simultaneously provide spatial discrimination. Thus, if a backbone is to preserve geometric compatibility while continuously renewing discriminative spatial information throughout the hierarchy, spatial interaction must be assigned to geometry-aware spatial mixing. The main generic learnable capacity and semantic recombination can then remain in pointwise channel mixing. From this perspective, the *MetaFormer-like separation* between spatial and channel mixing is a *structural trade-off* among invariance, discriminability, and scalable learning.

Guided by the principles and costs, we propose GeoFormer as a *unified architecture* for introducing geometric invariance priors into vision Transformers. It extends MetaFormer-like designs to a *richer spectrum* of invariance priors, from heuristic priors, to highly designed localized-similarity priors, and to fully data-adaptive convolution priors. GeoFormer allows us to achieve concrete trade-offs among invariance, discriminability, and scalability more flexibly in practical tasks, supported by substantial numerical evidence on CIFAR-100 and ImageNet.

**Index Terms**—Vision Transformers, geometric invariance, inductive priors.



## 1 INTRODUCTION

GEOMETRIC variability is one of the most persistent sources of complexity in vision. The same local visual structure may appear at different locations, orientations, and scales, while the semantic prediction is often expected to remain stable. *Invariance*—the property that a representation is stable under a family of transformations—has therefore served as a central inductive prior throughout the development of computer vision [1]. From a statistical perspective, invariance can be viewed as compressing *equivalence classes* induced by transformations [2], thereby reducing the need for exhaustive data coverage and mitigating the *curse of dimensionality*.

Yet in modern vision Transformers, the role of invariance has become increasingly ambiguous. The strongest Transformers often acquire invariance implicitly through large datasets, aggressive augmentation, and over-parameterization [3]. Conversely, making invariance explicit

by hard-wiring geometric constraints has been repeatedly criticized as being overly restrictive, potentially harming discriminability and scalability [4]. Therefore, this paper focuses on the role of geometric invariance priors in vision Transformers, from the perspective of historical review and principled characterization.

### 1.1 State of the Art and Motivation

Historical approaches to invariance in vision can be broadly organized into four routes, each making different trade-offs with discriminability and scalability.

#### 1.1.1 Hand-crafted Invariants

Classical computer vision treated invariance as an explicit design objective. Representative examples include moment invariants [5], [6] and local descriptors such as SIFT [7] and HOG [8], which were designed to remain stable under translation, rotation, scaling, and related geometric transformations.

The main limitation of this line is that the resulting features are usually fixed by design and often global, limiting spatial selectivity, adaptability, and integration with modern end-to-end hierarchical learning [9], [10].

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### 1.1.2 CNNs and Implicit Learning

CNNs replaced hand-crafted descriptors with learned hierarchical representations and established convolution as the dominant spatial inductive bias in vision. Relative to fully connected MLP-style models, convolution introduces locality, weight sharing, and translation equivariance, yielding substantially better statistical efficiency and practical performance on images [11], [12], [13], [14].

At the same time, the geometric bias of standard CNNs is concentrated mainly on translation, while robustness to broader transformations is usually acquired implicitly through data augmentation, large-scale supervision, and model capacity [15], [16].

### 1.1.3 Equivariant CNNs and Geometric Deep Learning

Equivariant networks were developed to combine the geometric explicitness of invariant methods with the hierarchical learning capacity of deep models. Rather than enforcing invariance directly, they preserve transformation structure across layers so that invariant representations can be derived in a principled way [4], [17], [18], [19], [20], [21].

Yet equivariance often comes with a specific structural commitment – geometry is *moved into* the channel dimension – channels commonly carry structured group representations to preserve group structure. This geometry–channel coupling can restrict generic cross-channel interaction and increase implementation complexity through specialized kernels, group discretization, and less hardware-friendly computation.

### 1.1.4 Transformers and Inductive Bias Reconsidered

Recent vision backbones reopened the question of how much explicit inductive bias is necessary. ViT [3], DeiT [22], and MLP-Mixer [23] showed that architectures with much weaker built-in spatial priors could still become highly competitive at sufficient scale. A common lesson, however, is that weaker architectural bias tends to place greater demands on data scale and training recipe, including augmentation and optimization [24], [25], [26].

Subsequent work moved toward more balanced designs by reintroducing lightweight spatial priors into otherwise generic backbones. Swin Transformer [27] restores locality and hierarchy through window-based attention, S2-MLP [28] introduces local spatial interaction through shift-based mixing, and ConvNeXt [29] reincorporates convolutional priors such as locality, hierarchy, and weight sharing. MetaFormer abstracts this trend into a general template: a vision backbone can be viewed as the decomposition of a spatial token mixer and a channel mixer, where the token mixer carries spatial interaction and the channel mixer performs generic semantic recombination [30].

### 1.1.5 Motivation

A more *principled characterization* of geometric invariance priors in vision Transformers remains lacking.

As shown in Table 1, the historical routes suggest that invariance, discriminability, and scalability are coupled through concrete trade-offs in vision models. Recent engineering attempts in vision Transformers further reinforce

TABLE 1

Driven by the invariance–discriminability–scalability tension, we characterize the principled form and structural cost of geometric invariance priors in vision Transformers, enabling our GeoFormer to achieve flexible, task-specific trade-offs among the three properties

|                  | Invariance | Discriminability | Scalability |
|------------------|------------|------------------|-------------|
| Invariants       | ✓          | ✗                | ✗           |
| Implicit CNNs    | —          | ✓                | ✓           |
| Equivariant CNNs | ✓          | ✓                | —           |
| Transformers     | ✗          | ✓                | ✓           |
| GeoFormer        | ✓          | ✓                | ✓           |

Marks: ✓ : built-in; — : partial; ✗ : not built-in.

this observation. Nevertheless, we still lack 1) a clear theoretical analysis of the proper form for introducing geometric invariance priors into vision Transformers, 2) the relationship between these possible forms and the historical forms of invariance in computer vision, and 3) the specific trade-off costs induced by such priors. We believe that this problem is important because it connects the historical development of computer vision with the design of future Transformer architectures.

## 1.2 Main Ideas and Contributions

Building on this motivation, we develop, to our knowledge, the first systematic characterization of the principled form and structural cost carried by geometric invariance priors in vision Transformers, which in turn gives a geometric interpretation and positioning to MetaFormer-like designs. Concretely, the contributions of this study are as follows.

- (1) **Formalization.** We formalize geometric compatibility via *localized Euclidean-similarity* transformations, capturing the intuition that a generic learning rule should not depend on the local position, orientation, or scale at which a visual structure happens to be sampled. This can be viewed as a localized form of the long-standing rotation–scale–translation (RST) hypothesis [7] in vision, and provides a starting point for our principled characterization. Moreover, we conduct our characterization under a *scalar* feature setting for generic and scalable learning, as in standard Transformers, rather than under the structured channel types in equivariant CNNs [4], [21].
- (2) **Theorem.** We prove the constraint theorem induced by this formulation: requiring localized-similarity compatibility with scalar-like channels forces any regular local generic learnable operator to collapse to *shared pointwise channel mixing*. In the linear case, this result reinterprets the ubiquitous  $1 \times 1$  convolution in vision networks. Beyond its familiar engineering roles in channel adjustment and efficient computation, it can also be viewed as the *principled form* of geometry-compatible generic learning in scalar-channel backbones.
- (3) **Division.** We derive the architectural consequence of this theorem. The pointwise collapse exposes a *spatial-blindness cost*: geometry-compatible generic learning cannot by itself provide spatial discrimination. Thus, if a Transformer is to preserve geometric compatibility while continuously renewing discriminative spatial information throughout the hierarchy, it should introduce

a separate mixer that is geometry-aware and confined to the spatial dimension. This gives a geometric interpretation of the *MetaFormer-like separation* [30]: a pervasive pattern in vision Transformers, previously justified mainly by empirical success, can be understood as a *structural trade-off* among invariance, discriminability, and scalable learning.

- (4) **Realization.** We instantiate *GeoFormer* by constructing geometry-aware spatial mixers from the classical theory of moment invariants [9], [10]. *GeoFormer* places geometric invariance priors in spatial mixing while preserving generic and scalable capacity in channel mixing. Controlled operator comparisons under matched frameworks show that the predicted spatial-blindness cost and role division are empirically observable, and that *GeoFormer* achieves a more favorable trade-off between invariance, discriminability, and scalability at ImageNet scale.

## 2 FORMALIZATIONS

To formalize geometric invariance priors, we fix three minimal ingredients. We first recall covariance, equivariance, and invariance as possible transformation laws in a representation hierarchy. We then model local visual geometry by localized Euclidean similarities, a local form of the classical rotation–scale–translation (RST) model in vision. Finally, we adopt a scalar-channel feature-space model for standard Transformers, in contrast to feature spaces with prescribed geometric channel types.

### 2.1 Covariance, Equivariance, and Invariance

Modern visual backbones, including vision Transformers, build hierarchical representations.

A geometric transformation acts on the input image, while *invariance* is typically a requirement at the output level. Between these two ends, the induced effect of the transformation should remain organized across intermediate feature spaces; otherwise, the hierarchy either discards useful spatial structure too early or lets geometry vary unpredictably.

*Equivariance* and *covariance* provide this formal control by requiring each layer to map the action on its input representation to the corresponding action on its output representation, so that an invariant readout can be applied at the end of the hierarchy [31].

**Definition 1** (Covariance, Equivariance, and Invariance). Let  $\mathfrak{G}$  be a geometric transformation family with generic element  $g$ . Let  $\mathcal{X}$  and  $\mathcal{Y}$  be representation spaces equipped with actions

$$\rho_{\mathcal{X}}(g) : \mathcal{X} \rightarrow \mathcal{X}, \quad \rho_{\mathcal{Y}}(g) : \mathcal{Y} \rightarrow \mathcal{Y}. \quad (1)$$

A representation map  $R : \mathcal{X} \rightarrow \mathcal{Y}$  is *covariant* with respect to  $(\rho_{\mathcal{X}}, \rho_{\mathcal{Y}})$  if

$$R(\rho_{\mathcal{X}}(g)X) = \rho_{\mathcal{Y}}(g)R(X), \quad X \in \mathcal{X}, g \in \mathfrak{G}. \quad (2)$$

When  $\mathcal{X}$  and  $\mathcal{Y}$  carry the same type of geometric action, this condition is called *equivariance*. When the output action is trivial,  $\rho_{\mathcal{Y}}(g) = \text{Id}$ , it becomes *invariance*:

$$R(\rho_{\mathcal{X}}(g)X) = R(X), \quad X \in \mathcal{X}, g \in \mathfrak{G}. \quad (3)$$

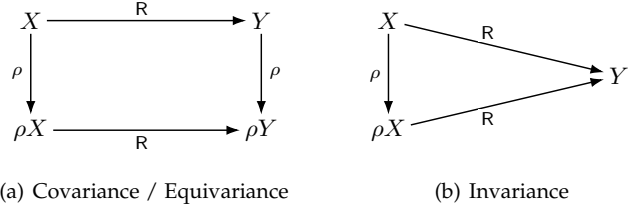


Fig. 1. Commutative diagrams for equivariance, covariance, and invariance. Here  $Y = R(X)$ . Equivariance and covariance correspond to the commutativity between the representation map  $R$  and the group action  $\rho$ , whereas invariance means that  $R$  is insensitive to the action of  $\rho$ .

Fig. 1 gives a compact commutative-diagram view of these transformation laws.

Thus, covariance is the general intertwining law between two representation spaces. Equivariance is the special case in which the input and output spaces carry the same kind of geometric action, and invariance is the special case in which the output action is trivial.

We now apply these notions to a feature hierarchy. Let  $\mathcal{X}_0, \dots, \mathcal{X}_L$  be representation spaces, each equipped with an action as (1):

$$\rho_l(g) : \mathcal{X}_l \rightarrow \mathcal{X}_l, \quad g \in \mathfrak{G}, \quad (4)$$

and define the hierarchy as layer maps

$$F_{l+1} = \Phi_l(F_l), \Phi_l : \mathcal{X}_l \rightarrow \mathcal{X}_{l+1}, l = 0, \dots, L-1. \quad (5)$$

Controlled propagation through intermediate layers is expressed by the layerwise covariance condition as (2):

$$\Phi_l(\rho_l(g)F_l) = \rho_{l+1}(g)\Phi_l(F_l), \quad g \in \mathfrak{G}. \quad (6)$$

When consecutive representation spaces carry the same type of geometric action, (6) is the usual equivariance condition.

After such controlled propagation, task-level invariance can be imposed by a readout  $\Psi : \mathcal{X}_L \rightarrow \mathcal{Y}$  whose output action is trivial as (3):

$$\Psi(\rho_L(g)F_L) = \Psi(F_L), \quad g \in \mathfrak{G}. \quad (7)$$

In a theoretical blueprint [31], when a geometric transformation acts on the input image, equivariant/covariant layers can manage geometric effects through the representation hierarchy (6), so that the final readout can eliminate the effects for task-level invariance (7).

### 2.2 Localized Euclidean Similarity

The previous subsection states transformation laws for an abstract transformation family  $\mathfrak{G}$ . We now instantiate this family by the local coordinate geometry of visual structures.

Such variability shows up as changes in position, orientation, and scale: the local coordinate frame in which a structure is sampled shifts, while the relative spatial arrangement that defines the structure itself is preserved. This rotation–scale–translation (RST) geometry is the classical basis of scale-space theory and SIFT-like local invariant features [7], [32].

It must be modeled locally because viewpoint change and object motion generally induce spatially varying image deformations: a single global transformation cannot account

for all local structures, whereas simple geometric laws remain meaningful on local regions [33].

Among local models, Euclidean similarity retains the RST variables essential to visual invariance while avoiding the unconstrained freedom of arbitrary deformation. We therefore model visual geometry by localized Euclidean similarities, namely local RST frame changes smoothly embedded into the full spatial domain.

**Definition 2** (Localized Euclidean Similarity). *Let  $\Omega \subset \mathbb{R}^2$  be a spatial domain. We denote by  $\mathfrak{G}_{\text{loc-sim}}(\Omega)$  the set of diffeomorphisms  $g : \Omega \rightarrow \Omega$  for which there exist open sets  $U \Subset V \Subset \Omega$ , a translation  $a \in \mathbb{R}^2$ , a scale  $s > 0$ , and a rotation  $Q \in \text{SO}(2)$ , such that, with*

$$h(\xi) = a + sQ\xi, \quad (8)$$

one has

$$\begin{aligned} U \cup h(U) &\Subset V, \\ g|_U &= h|_U, \quad g|_{\Omega \setminus V} = \text{Id}. \end{aligned} \quad (9)$$

Any  $g \in \mathfrak{G}_{\text{loc-sim}}(\Omega)$  is called a *localized Euclidean-similarity transformation*.

Here  $h$  is the local RST coordinate change, while  $g$  is its full-domain localization. The set  $U$  is the active region on which the coordinate change is exact, and  $V$  is the support region in which this change is smoothly embedded into  $\Omega$ . Thus  $\mathfrak{G}_{\text{loc-sim}}(\Omega)$  retains the classical RST geometry on each active region while allowing the visual coordinate frame to vary across space. In what follows, unless otherwise specified, the abstract family  $\mathfrak{G}$  in the transformation laws above refers to  $\mathfrak{G}_{\text{loc-sim}}(\Omega)$ .

### 2.3 Scalar-Channel Feature Field

The previous subsection fixes the geometric family  $\mathfrak{G}_{\text{loc-sim}}(\Omega)$ . We now specify the feature spaces on which this family acts.

A feature field has two components: a spatial coordinate and a channel value. Image geometry canonically acts on the spatial coordinate. An action on the channel value is not determined by image geometry itself; it requires prescribed geometric channel types. Standard vision Transformers avoid this additional commitment. Their channels are untyped learned coordinates, which keeps channel mixing generic and allows width to scale without preserving fixed representation-type blocks. We therefore use the scalar-channel model: geometry acts on space, while the channel value space carries the trivial action.

**Definition 3** (Scalar-Channel Feature Field). *Let  $\Omega \subset \mathbb{R}^2$  be a spatial domain and let  $C \in \mathbb{N}$ . The scalar-channel feature space of width  $C$  over  $\Omega$  is*

$$\mathcal{F}_C(\Omega) = \{F : \Omega \rightarrow \mathbb{R}^C\}. \quad (10)$$

For  $g \in \mathfrak{G}_{\text{loc-sim}}(\Omega)$ , the scalar-channel action  $\rho_{\mathcal{F}_C}(g) : \mathcal{F}_C(\Omega) \rightarrow \mathcal{F}_C(\Omega)$  is defined by

$$[\rho_{\mathcal{F}_C}(g)F](\xi) = F(g^{-1}\xi), \quad \xi \in \Omega. \quad (11)$$

An element of  $\mathcal{F}_C(\Omega)$  equipped with this action is called a *scalar-channel feature field*.

At layer  $l$ , a Transformer token array is modeled as

$$F_l \in \mathcal{F}_{C_l}(\Omega_l), \quad (12)$$

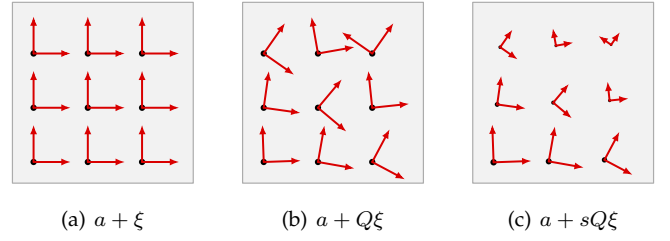


Fig. 2. Visualization of localized similarity modeling. From left to right, the panels illustrate how the translation parameter  $a$ , rotation parameter  $Q$ , and scale parameter  $s$  affect local coordinate frames. This model reflects a basic spatial pattern of natural images and follows the classical rotation–scale–translation assumption in computer vision.

where  $\Omega_l$  is the spatial token domain and  $C_l$  is the channel width. In the notation of Definition 1, this means  $\mathcal{X}_l = \mathcal{F}_{C_l}(\Omega_l)$ . Thus a geometric transformation moves where feature values are sampled, but does not rotate, permute, or mix the channel basis.

Note that structured equivariant models make a different feature-space choice. They use a nontrivial fiber action, which can be written as  $[\rho^\tau(g)F](\xi) = \tau(g, \xi)F(g^{-1}\xi)$  with  $\tau(g, \xi) \neq \text{Id}$  – the channels are assigned by prescribed geometric types [4], [21]. The scalar-channel model is the minimal case  $\tau(g, \xi) = \text{Id}$ . It is therefore the least committed feature-space model aligned with standard vision Transformer practices [3], [22], [23], [27], [30].

## 3 THEORETICAL PRINCIPLES AND COSTS

We now turn the invariance–discriminability–scalability tension into an operator-level characterization. This section derives the principled form and structural cost of geometry-compatible learning from the formalizations in Sec. 2.

We first show that requiring localized-similarity compatibility with scalar-like channels forces any regular local generic learnable operator to collapse to shared pointwise channel mixing. We then show that this form is spatially blind, which forces a separate spatial role whenever the hierarchy must renew spatial discrimination. Finally, we formalize the resulting token–channel role division: the spatial component determines the geometric prior or structural constraint imposed on the hierarchy, while the pointwise channel component preserves and recombines that structure.

### 3.1 The Pointwise-Collapse Principle

We now specialize the formalizations in Sec. 2 to regular local learnable operators of standard vision Transformers, and classify which operators can satisfy compatibility under the geometric and feature-space models.

**Theorem 1** (Pointwise-Collapse Principle). *Let  $\Omega \subset \mathbb{R}^2$  be connected. Under the localized Euclidean-similarity model in Definition 2 and the scalar-channel feature model in Definition 3, let the feature operator  $\mathsf{T} : \mathcal{F}_{C_{\text{in}}}(\Omega) \rightarrow \mathcal{F}_{C_{\text{out}}}(\Omega)$  be continuous and regular local. Here regular locality means that, for every  $\xi \in \Omega$ , there exist an integer  $q \geq 0$  and a continuous map  $P_\xi$  such that*

$$\mathsf{T}(F)(\xi) = P_\xi(j_\xi^q F), \quad (13)$$

where

$$j_\xi^q F = (\partial^\alpha F(\xi))_{|\alpha| \leq q} \quad (14)$$

is the finite jet of  $F$  at  $\xi$ . Then the following two statements are equivalent.

(i) Localized-similarity compatibility. The feature operator  $T$  satisfies the layerwise covariance law (6) specialized to scalar-channel feature fields:

$$T(\rho_{\mathcal{F}_{C_{in}}}(g)F) = \rho_{\mathcal{F}_{C_{out}}}(g)T(F), \quad g \in \mathfrak{G}_{\text{loc-sim}}(\Omega). \quad (15)$$

(ii) Shared pointwise channel mixing. There exists a continuous channel map  $c : \mathbb{R}^{C_{in}} \rightarrow \mathbb{R}^{C_{out}}$  such that

$$T(F)(\xi) = c(F(\xi)), \quad F \in \mathcal{F}_{C_{in}}(\Omega), \xi \in \Omega. \quad (16)$$

Equivalently, within continuous regular local scalar-channel operators, localized-similarity compatibility selects exactly shared pointwise channel mixing.

*Proof.* The implication (ii) $\Rightarrow$ (i) is immediate from the scalar-channel action. Indeed, if  $T(F)(\xi) = c(F(\xi))$ , then

$$\begin{aligned} T(\rho_{\mathcal{F}_{C_{in}}}(g)F)(\xi) &= c(F(g^{-1}\xi)) \\ &= [\rho_{\mathcal{F}_{C_{out}}}(g)T(F)](\xi). \end{aligned} \quad (17)$$

We prove (i) $\Rightarrow$ (ii) in two steps. First, by regular locality, around any point  $\xi_0 \in \Omega$ , one can write

$$T(F)(\xi_0) = P_{\xi_0}(F(\xi_0), (\partial^\alpha F(\xi_0))_{1 \leq |\alpha| \leq q}), \quad (18)$$

where  $P_{\xi_0}$  is continuous in the jet variables.

Choose a localized isotropic rescaling  $g_s \in \mathfrak{G}_{\text{loc-sim}}(\Omega)$  that fixes  $\xi_0$  and agrees near  $\xi_0$  with

$$g_s(\xi) = \xi_0 + s(\xi - \xi_0), \quad s > 0. \quad (19)$$

Because  $g_s(\xi_0) = \xi_0$ , compatibility gives

$$T(\rho_{\mathcal{F}_{C_{in}}}(g_s)F)(\xi_0) = T(F)(\xi_0). \quad (20)$$

Since  $g_s^{-1}(\xi) = \xi_0 + s^{-1}(\xi - \xi_0)$  near  $\xi_0$ , the positive-order jets transform as

$$\partial^\alpha (\rho_{\mathcal{F}_{C_{in}}}(g_s)F)(\xi_0) = s^{-|\alpha|} \partial^\alpha F(\xi_0), \quad |\alpha| \geq 1. \quad (21)$$

Substituting (21) into (18) and using (20) yields

$$\begin{aligned} P_{\xi_0}(F(\xi_0), (s^{-|\alpha|} \partial^\alpha F(\xi_0))_{1 \leq |\alpha| \leq q}) \\ = P_{\xi_0}(F(\xi_0), (\partial^\alpha F(\xi_0))_{1 \leq |\alpha| \leq q}). \end{aligned} \quad (22)$$

This identity holds for arbitrary local jets of  $F$ . Letting  $s \rightarrow \infty$  and using the continuity of  $P_{\xi_0}$ , all positive-order jet variables vanish:

$$P_{\xi_0}(F(\xi_0), (\partial^\alpha F(\xi_0))_{1 \leq |\alpha| \leq q}) = P_{\xi_0}(F(\xi_0), 0). \quad (23)$$

Thus  $T$  reduces to a zeroth-order operator:

$$T(F)(\xi) = c_\xi(F(\xi)) \quad (24)$$

for a point-dependent channel map  $c_\xi$ .

Second, localized Euclidean similarities contain local transports between neighborhoods. For any  $\xi, \eta \in \Omega$ , choose  $g \in \mathfrak{G}_{\text{loc-sim}}(\Omega)$  with  $g(\xi) = \eta$ . Evaluating compatibility at  $\eta$  and using (24) gives

$$c_\eta(F(\xi)) = c_\xi(F(\xi)). \quad (25)$$

Since  $F(\xi) \in \mathbb{R}^{C_{in}}$  is arbitrary,  $c_\eta = c_\xi$ . The connectedness of  $\Omega$  then implies  $c_\xi \equiv c$ , which proves (16).  $\square$

Theorem 1 should be understood as a very strong theoretical result on selectivity:

*Even when a classical model of visual geometry is imposed on standard scalar-channel Transformer feature spaces, compatibility removes all positive-order spatial jet degrees of freedom. The remaining generic learnable freedom lies in the shared pointwise channel mixing.*

**Remark 1** (Minimal Collapse Mechanism). *The collapse does not require the full RST content of localized Euclidean similarity. Only localized isotropic rescaling and localized transport are used: rescaling eliminates positive-order spatial jet dependence, and transport makes the remaining pointwise channel map spatially shared.*

With Remark 1, we note that rotation in the classical RST model of Definition 2 is not even part of the collapse mechanism. This makes the theorem stronger than it literally suggests: a smaller compatibility requirement than the complete local RST model is already enough to force pointwise collapse. Consequently, in the standard Transformer regime where channels remain scalable and untyped, a localized-geometry-compatible regular local operator has almost no spatial design freedom: its admissible form is a shared pointwise channel map.

**Remark 2** (Discrete Realization). *On a regular token grid, the linear special case  $T(F)(\xi) = WF(\xi)$  is exactly shared  $1 \times 1$  convolution. The general form  $T(F)(\xi) = c(F(\xi))$  corresponds to a shared pointwise channel mixer, such as a pointwise MLP.*

With Remark 2, we note that there is another interpretation of the widespread use of  $1 \times 1$  convolution and pointwise channel MLPs, going beyond familiar engineering interpretations such as bottleneck designs and efficient computation [13], [14], [34]. Theorem 1 adds a geometric motivation: shared pointwise channel mixing is precisely the clean form selected by localized geometric compatibility – it preserves the geometric action while retaining generic channel recombination.

### 3.2 The Spatial-Blindness Cost

Theorem 1 serves as a theoretical filter: it singles out the clean form compatible with localized visual geometry and generic scalable learning. This selected form is also pervasive in practice, realized as shared  $1 \times 1$  convolution in the linear case and as pointwise channel MLPs in the nonlinear case.

It is therefore natural to ask whether a Transformer hierarchy can be built solely from this clean compatible operator. We next formulate its direct structural cost: spatial blindness. Before that, we need to introduce the following notion.

**Definition 4** (Local Spatial Discrimination). *Let  $O : \mathcal{F}_{C_{in}}(\Omega) \rightarrow \mathcal{F}_{C_{out}}(\Omega)$  be a feature operator. We say that  $O$  realizes local spatial discrimination if there exist a point  $\xi \in \Omega$ , a neighborhood  $U_\xi \subset \Omega$  of  $\xi$ , and two feature fields  $F, \tilde{F} \in \mathcal{F}_{C_{in}}(\Omega)$  such that*

$$F(\xi) = \tilde{F}(\xi), F|_{\Omega \setminus U_\xi} = \tilde{F}|_{\Omega \setminus U_\xi}, F|_{U_\xi} \neq \tilde{F}|_{U_\xi}, \quad (26)$$

but

$$O(F)(\xi) \neq O(\tilde{F})(\xi). \quad (27)$$

In this case, the output at  $\xi$  distinguishes two feature fields that agree at  $\xi$  but differ in the local neighborhood of  $\xi$ .

In Definition 4, we specify the minimal discrimination requirement: when two feature fields have the same value at a center point, an operator with spatial discrimination should be able to distinguish them if their neighborhoods around that point differ.

**Corollary 1** (Spatial-Blindness Cost). *Let  $T_l : \mathcal{F}_{C_{l-1}}(\Omega) \rightarrow \mathcal{F}_{C_l}(\Omega)$ ,  $l = 1, \dots, L$ , be admissible regular local learnable operators under Theorem 1. Then any composition*

$$B = T_L \circ \dots \circ T_1 : \mathcal{F}_{C_0}(\Omega) \rightarrow \mathcal{F}_{C_L}(\Omega) \quad (28)$$

*cannot realize local spatial discrimination in the sense of Definition 4.*

*Proof.* By Theorem 1, each admissible operator has the shared pointwise form

$$T_l(Z)(\xi) = c_l(Z(\xi)), Z \in \mathcal{F}_{C_{l-1}}(\Omega), l = 1, \dots, L, \quad (29)$$

for some channel map  $c_l : \mathbb{R}^{C_{l-1}} \rightarrow \mathbb{R}^{C_l}$ . Therefore the whole composition remains pointwise:

$$B(F)(\xi) = (c_L \circ \dots \circ c_1)(F(\xi)). \quad (30)$$

Now take any  $F, \tilde{F}, \xi, U_\xi$  satisfying the neighborhood-difference condition in (26). In particular,  $F(\xi) = \tilde{F}(\xi)$ . Hence (30) gives

$$\begin{aligned} B(F)(\xi) &= (c_L \circ \dots \circ c_1)(F(\xi)) \\ &= (c_L \circ \dots \circ c_1)(\tilde{F}(\xi)) = B(\tilde{F})(\xi). \end{aligned} \quad (31)$$

Thus  $B$  cannot satisfy (27). Therefore it cannot realize local spatial discrimination.  $\square$

As shown in Fig. 3, this paper has revealed the internal tension among invariance, discriminability, and scalability in vision Transformers. This tension has been perceived in past engineering practice (Table 1), but has only now been given a clear theoretical characterization.

The above characterization is not merely abstract: an important example shows how it exposes the limitation of a concrete architectural pattern.

**Example 1** (Limitation of the Front-End/Back-End Split). *An intuitive way to embed invariance into a visual network with discriminability and scalability is to use hand-crafted invariant operators to extract front-end features, and then feed them into a generic learnable back-end [35]. Under our theoretical characterization, the key issue is whether this generic back-end can respect the geometric behavior of the front-end features. The conclusion is restrictive: this requirement forces the back-end toward pointwise channel operators, so the back-end cannot renew local spatial discrimination.*

This front-end/back-end pattern appears in wavelet scattering networks [35], a mathematical invariance framework with wavelet front-end and pointwise back-end. Our characterization interprets its architectural design as a trilemma.

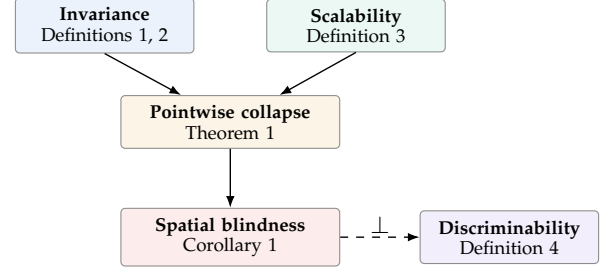


Fig. 3. Theoretical characterization of the invariance–discriminability–scalability tension. Invariance, modeled by Definitions 1 and 2, together with scalability, modeled by Definition 3, leads through compatibility analysis to the pointwise-collapse theorem. The induced spatial-blindness cost then stands in direct conflict with discriminability, modeled by Definition 4. This theoretical characterization echoes the practical observation summarized in Table 1.

### 3.3 Spatial–Channel Role Division

This tension calls for a more refined architectural pattern, beyond heuristic designs such as Example 1. In this architectural pattern, there should be a separate spatial operator to close the spatial-discrimination gap. Moreover, the front-end/back-end lesson suggests that this spatial operator should be distributed throughout the Transformer hierarchy, so that spatial discrimination can be continuously renewed across layers.

Up to this point, readers familiar with vision Transformers may recognize the connection to current architectural practices. The dominant learnable capacity is often assigned to pointwise channel operators, while additional spatial operators, such as pooling or convolutional mixers, introduce spatial priors. Here, MetaFormer summarizes such practices by treating token mixing and channel mixing as two basic ingredients for designing vision backbones [30].

We next formalize what this MetaFormer-like spatial–channel role division means under our characterization of the invariance–discriminability–scalability tension.

**Definition 5** (Spatial–Channel Role Division). *A layer  $\Phi_l : \mathcal{F}_{C_l}(\Omega) \rightarrow \mathcal{F}_{C_{l+1}}(\Omega)$  is said to follow the spatial–channel role division if it admits the decomposition*

$$\Phi_l = C_l \circ S_l. \quad (32)$$

*The spatial component  $S_l : \mathcal{F}_{C_l}(\Omega) \rightarrow \mathcal{F}_{\tilde{C}_l}(\Omega)$  is channel-separable: writing  $F = (F_1, \dots, F_{C_l})$ , there exist spatial operators  $\varphi_{l,i} : \mathcal{F}_1(\Omega) \rightarrow \mathcal{F}_{r_{l,i}}(\Omega)$  such that*

$$S_l(F) = (\varphi_{l,1}(F_1), \dots, \varphi_{l,C_l}(F_{C_l})), \quad \tilde{C}_l = \sum_{i=1}^{C_l} r_{l,i}. \quad (33)$$

*The channel component  $C_l : \mathcal{F}_{\tilde{C}_l}(\Omega) \rightarrow \mathcal{F}_{C_{l+1}}(\Omega)$  is shared and pointwise:*

$$C_l(Z)(\xi) = c_l(Z(\xi)), \quad Z \in \mathcal{F}_{\tilde{C}_l}(\Omega), \quad \xi \in \Omega, \quad (34)$$

*where  $c_l : \mathbb{R}^{\tilde{C}_l} \rightarrow \mathbb{R}^{C_{l+1}}$  is independent of  $\xi$ .*

Note that  $C_l$  and  $S_l$  are orthogonal in feature space. Technically, this is similar to depthwise separable convolution [14]. Definition 5 distinguishes the spatial–channel role division from standard convolution in CNNs, where spatial and channel mixing are jointly performed by the same convolutional kernel [11]. Such a separation leads to a

clean property in the hierarchical geometric analysis below: the geometric prior carried by the hierarchical features is entirely controlled by  $S_l$ , while  $C_l$  does not break this prior throughout the hierarchy.

**Proposition 1** (Hierarchical Propagation of Geometric Priors). *Let  $\mathfrak{H}$  be a family of invertible spatial transformations on  $\Omega$ . For each  $\gamma \in \mathfrak{H}$  and each channel width  $C$ ,  $\rho_{\mathcal{F}_C}(\gamma)$  denotes the scalar-channel pullback action on  $\mathcal{F}_C(\Omega)$ , following the convention of (11). Let  $\{\Phi_l\}_{l=0}^{L-1}$  be layers following the spatial-channel role division in Definition 5, and let*

$$P_k = \Phi_{k-1} \circ \dots \circ \Phi_0, \quad k = 1, \dots, L, \quad (35)$$

*be the prefix cascade. Assume that every spatial component  $S_l$  imposes the same  $\mathfrak{H}$ -covariance, namely, for every  $F_l \in \mathcal{F}_{C_l}(\Omega)$  and every  $\gamma \in \mathfrak{H}$ ,*

$$S_l(\rho_{\mathcal{F}_{C_l}}(\gamma)F_l) = \rho_{\mathcal{F}_{C_l}}(\gamma)S_l(F_l). \quad (36)$$

*Then, for every  $k = 1, \dots, L$ , every  $F_0 \in \mathcal{F}_{C_0}(\Omega)$ , and every  $\gamma \in \mathfrak{H}$ , the prefix cascade preserves the same  $\mathfrak{H}$ -covariance:*

$$P_k(\rho_{\mathcal{F}_{C_0}}(\gamma)F_0) = \rho_{\mathcal{F}_{C_k}}(\gamma)P_k(F_0). \quad (37)$$

*Proof.* For every  $Z \in \mathcal{F}_{\tilde{C}_l}(\Omega)$  and every  $\gamma \in \mathfrak{H}$ , the pointwise channel component satisfies

$$C_l(\rho_{\mathcal{F}_{\tilde{C}_l}}(\gamma)Z) = \rho_{\mathcal{F}_{C_{l+1}}}(\gamma)C_l(Z). \quad (38)$$

This follows directly from  $C_l(Z)(\xi) = c_l(Z(\xi))$ . Combining (36) with (38) gives, for every  $F_l \in \mathcal{F}_{C_l}(\Omega)$  and every  $\gamma \in \mathfrak{H}$ ,

$$\Phi_l(\rho_{\mathcal{F}_{C_l}}(\gamma)F_l) = \rho_{\mathcal{F}_{C_{l+1}}}(\gamma)\Phi_l(F_l). \quad (39)$$

Applying (39) inductively along  $P_k = \Phi_{k-1} \circ \dots \circ \Phi_0$  proves (37).  $\square$

Proposition 1 shows that the family  $\mathfrak{H}$  specifies not only the spatial prior imposed at a *single* layer, but also the same prior carried through a *hierarchy* following the spatial-channel role division.

Therefore, in practice, the geometric prior of such a hierarchical representation reduces to the choice of the family  $\mathfrak{H}$  and the realization of the spatial component  $S_l$ . Naturally, there are four potential choices at different levels, some of which have already become mature components in vision Transformers.

**Example 2** (Representative Levels of Geometric Priors). *Four representative choices are*

$$\begin{aligned} \mathfrak{H}_{\text{none}} &= \{\text{Id}\}, & \mathfrak{H}_{\text{tr}} &= \text{Trans}_{\text{grid}}, \\ \mathfrak{H}_{\text{sq}} &= \text{Trans}_{\text{grid}} \rtimes D_4, & \mathfrak{H}_{\text{loc-sim}} &= \mathfrak{G}_{\text{loc-sim}}. \end{aligned} \quad (40)$$

Here  $\text{Trans}_{\text{grid}}$  denotes grid translations, and  $D_4$  denotes the square-grid symmetries generated by  $90^\circ$  rotations and flips. The four choices correspond, respectively, to weak-prior ViT/MLP-style token mixing [3], [23], convolutional token mixing [29], [36], pooling-type token mixing such as PoolFormer [30], and localized-RST spatial mixing based on local invariant geometry [9], [10]. The last choice is the theoretical position of our **GeoFormer**, whose realization will be developed in the next section.

We can see that all the designs are already included under our unified theoretical umbrella: their geometric motivations and possible extensions (e.g., **GeoFormer**) become transparent.

## 4 GEOFORMER

After the operator characterization in Sec. 3, we have obtained a new understanding of the principles and costs of invariance priors in vision Transformers. More importantly, it guides our search for concrete *architectural* trade-offs among invariance, discriminability, and scalability.

This section presents **GeoFormer** as a concrete realization of our theory. It fills the key missing position in Example 2: the  $\mathfrak{H}_{\text{loc-sim}} = \mathfrak{G}_{\text{loc-sim}}$  level. In this sense, **GeoFormer** extends the architectural spectrum of invariance priors in vision Transformers, thereby providing more choices for concrete trade-offs.

We first use invariant theory to construct a spatial mixer that is  $\mathfrak{G}_{\text{loc-sim}}$ -covariant. This spatial mixer is then combined with pointwise channel mixers to form a block, the core learning component of a MetaFormer-like architecture, whose geometric prior remains consistent with  $\mathfrak{G}_{\text{loc-sim}}$ . Correspondingly, the geometric covariance propagated by compositions of such blocks can be converted into a global invariant by a  $\mathfrak{G}_{\text{loc-sim}}$ -invariant readout, following the blueprint in Sec. 2.1. Finally, we specify how these blocks form **GeoFormer** and its variant **GAFormer**, and in particular introduce other engineering roles in the concrete trade-off<sup>1</sup>.

### 4.1 $\mathfrak{G}_{\text{loc-sim}}$ -Mixer

**GeoFormer** generalizes MetaFormer at the level of geometric invariance priors. Its core idea is to make the design of spatial mixers principled, and to place the geometric analysis of the entire architecture under a unified theoretical framework.

In this subsection, we take  $\mathfrak{G}_{\text{loc-sim}}$  as the geometric prior for the spatial mixer in **GeoFormer**. At the same time, it is important to keep the generality of **GeoFormer** in mind: **GeoFormer** is not a single concrete architecture. Rather, from the perspective of completing the existing spectrum of geometric priors, we choose  $\mathfrak{G}_{\text{loc-sim}}$  as the representative realization of **GeoFormer** in this paper.

To realize such a mixer, we largely follow our previous studies on local and hierarchical invariants [9], [10]. This gives a hand-crafted spatial operator with transparent  $\mathfrak{G}_{\text{loc-sim}}$ -covariance.

**Definition 6** ( $\mathfrak{G}_{\text{loc-sim}}$ -Mixer). *Let  $F \in \mathcal{F}_C(\Omega)$ , and let  $\Lambda$  be a finite ordered set of branch indices  $\lambda = (n, m, w)$ , where  $n$  is a radial order,  $m \in \mathbb{Z}$  is an angular order, and  $w > 0$  is the support radius. For  $\xi \in \Omega$ , write  $B(\xi, w) = \{\eta \in \Omega : \|\eta - \xi\| \leq w\}$ . For a scalar field  $f \in \mathcal{F}_1(\Omega)$ , define*

$$V_\lambda^\xi(\eta) = R_n \left( \frac{\|\eta - \xi\|}{w} \right) \exp(jm\theta_\xi(\eta)), \eta \in B(\xi, w), \quad (41)$$

where  $R_n$  is the chosen radial basis,  $j^2 = -1$ , and  $\theta_\xi(\eta) = \text{Arg}((\eta_1 - \xi_1) + j(\eta_2 - \xi_2))$ . The local response is

$$\nu_\lambda[f](\xi) = \frac{1}{w^2} \int_{B(\xi, w)} (V_\lambda^\xi(\eta))^* f(\eta) d\eta. \quad (42)$$

For each  $\lambda \in \Lambda$ , the scalar branch is

$$M^\lambda(f)(\xi) = |\nu_\lambda[f](\xi)|. \quad (43)$$

1. Code available at <https://github.com/ShurenQi/GeoFormer>.

For  $F = (F_1, \dots, F_C)$ , its channelwise lifting is

$$M_{\text{cw}}^\lambda(F) = (M^\lambda(F_1), \dots, M^\lambda(F_C)). \quad (44)$$

The corresponding mixer is

$$S_\Lambda(F) = \text{Concat}_{\lambda \in \Lambda} M_{\text{cw}}^\lambda(F). \quad (45)$$

**Proposition 2** ( $\mathfrak{G}_{\text{loc-sim}}$ -Mixer Covariance). *Let  $g \in \mathfrak{G}_{\text{loc-sim}}(\Omega)$ , and assume that on the local supports involved below,  $g$  agrees with the local similarity  $\ell_g(\eta) = a + sQ\eta$ , where  $s > 0$  and  $Q \in \text{SO}(2)$ . For  $\lambda = (n, m, w)$ , define*

$$\lambda^g = (n, m, w/s), \quad \Lambda^g = \{\lambda^g : \lambda \in \Lambda\}, \quad (46)$$

with the order induced from  $\Lambda$ . Then, whenever the transported local supports remain in the active similarity region,

$$S_\Lambda(\rho_{\mathcal{F}_C}(g)F) = \rho_{\mathcal{F}_{C|\Lambda}}(g)S_{\Lambda^g}(F). \quad (47)$$

*Proof.* We first establish the scalar branch transformation. Fix  $\xi \in \Omega$  such that the local support at  $\xi$  and its inverse image are contained in the active similarity region. Set  $\bar{\xi} = \ell_g^{-1}\xi = g^{-1}\xi$ , and change variables by  $\eta = \ell_g(\zeta)$ . Then  $d\eta = s^2 d\zeta$ ,  $B(\xi, w)$  is mapped to  $B(\bar{\xi}, w/s)$ , and

$$\frac{\|\eta - \xi\|}{w} = \frac{\|\zeta - \bar{\xi}\|}{w/s}. \quad (48)$$

If  $\alpha_Q$  is the rotation angle of  $Q$ , then  $\theta_\xi(\eta) = \theta_{\bar{\xi}}(\zeta) + \alpha_Q$  modulo  $2\pi$ . Hence

$$V_\lambda^\xi(\ell_g(\zeta)) = \exp(jm\alpha_Q)V_{\lambda^g}^{\bar{\xi}}(\zeta). \quad (49)$$

Using also  $[\rho_{\mathcal{F}_1}(g)f](\ell_g(\zeta)) = f(\zeta)$ , we obtain

$$\begin{aligned} \nu_\lambda[\rho_{\mathcal{F}_1}(g)f](\xi) &= \frac{1}{w^2} \int_{B(\xi, w)} (V_\lambda^\xi(\eta))^* f(g^{-1}\eta) d\eta \\ &= \exp(-jm\alpha_Q) \frac{1}{(w/s)^2} \int_{B(\bar{\xi}, w/s)} (V_{\lambda^g}^{\bar{\xi}}(\zeta))^* f(\zeta) d\zeta \\ &= \exp(-jm\alpha_Q) \nu_{\lambda^g}[f](g^{-1}\xi). \end{aligned} \quad (50)$$

For each  $\lambda \in \Lambda$ , taking magnitudes removes the unimodular phase:

$$M^\lambda(\rho_{\mathcal{F}_1}(g)f) = \rho_{\mathcal{F}_1}(g)M^{\lambda^g}(f). \quad (51)$$

Applying (51) to every channel of  $F = (F_1, \dots, F_C)$  gives

$$M_{\text{cw}}^\lambda(\rho_{\mathcal{F}_C}(g)F) = \rho_{\mathcal{F}_C}(g)M_{\text{cw}}^{\lambda^g}(F). \quad (52)$$

Concatenating (52) over the ordered family  $\Lambda$  proves (47).  $\square$

## 4.2 $\mathfrak{G}_{\text{loc-sim}}$ -Block

As the main learnable component of GeoFormer, the  $\mathfrak{G}_{\text{loc-sim}}$ -Mixer is repeatedly combined with pointwise channel mixers to form MetaFormer-like blocks. As anticipated by Proposition 1, such blocks inherit the geometric transformation law carried by the  $\mathfrak{G}_{\text{loc-sim}}$ -Mixer.

**Definition 7** ( $\mathfrak{G}_{\text{loc-sim}}$ -Block). *Let  $\star \in \{\text{b}, \text{ib}\}$ . A GeoFormer block is an operator  $\text{Blk}_\Lambda^\star : \mathcal{F}_C(\Omega) \rightarrow \mathcal{F}_C(\Omega)$  of the form*

$$\text{Blk}_\Lambda^\star = C_{\text{out}}^\star \circ S_\Lambda^\star \circ C_{\text{in}}^\star. \quad (53)$$

Here  $C_{\text{in}}^\star$  and  $C_{\text{out}}^\star$  are shared pointwise channel components in the sense of Definition 5, and  $S_\Lambda^\star$  is instantiated by either Example 3 or Example 4.

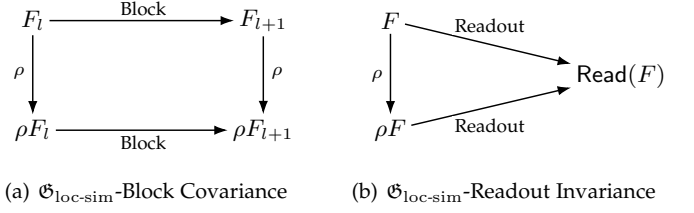


Fig. 4. Commutative diagrams for GeoFormer. The  $\mathfrak{G}_{\text{loc-sim}}$ -Block serves as the core learning component that propagates localized-similarity covariance, while the final  $\mathfrak{G}_{\text{loc-sim}}$ -Readout factors out the covariant behavior to produce global invariants. This engineering realization provides the architectural counterpart to the formal blueprint shown in Fig. 1.

Note that Definition 7 does not adopt the minimal block form implied by Proposition 1. This choice is mainly motivated by engineering trade-offs: both examples below have widely used architectural counterparts.

For the two block realizations below, let  $\lambda_0 = (0, 0, w) \in \Lambda$ . The response indexed by  $\lambda_0$  requires no magnitude and has much higher energy than the other components, it is handled separately for computational efficiency and energy balance. Set  $\Lambda_{\text{detail}} = \Lambda \setminus \{\lambda_0\}$ . For an intermediate feature  $Z$ , define the full-channel zero-order response component-wise by

$$[A_{\lambda_0}(Z)]_c = \nu_{\lambda_0}[Z_c]. \quad (54)$$

**Example 3** (Bottleneck). *In the bottleneck form,  $C_{\text{in}}^{\text{b}}$  maps  $\mathcal{F}_C(\Omega)$  to  $\mathcal{F}_{C_{\text{b}}}(\Omega)$ , with  $C_{\text{b}} < C$ . The full detail family is applied to every bottleneck channel:*

$$D_{\Lambda_{\text{detail}}}^{\text{b}}(Z) = \text{Concat}_{\lambda \in \Lambda_{\text{detail}}} M_{\text{cw}}^\lambda(Z). \quad (55)$$

The spatial mixer is

$$S_\Lambda^{\text{b}}(Z) = \text{Concat} [A_{\lambda_0}(Z), D_{\Lambda_{\text{detail}}}^{\text{b}}(Z)], Z \in \mathcal{F}_{C_{\text{b}}}(\Omega). \quad (56)$$

This form gives dense geometric exposure in the reduced channel space. Its cost is the information bottleneck introduced before spatial mixing.

**Example 4** (Inverted Bottleneck). *In the inverted-bottleneck form,  $C_{\text{in}}^{\text{ib}}$  maps  $\mathcal{F}_C(\Omega)$  to  $\mathcal{F}_{C_{\text{ib}}}(\Omega)$ , with  $C_{\text{ib}} \geq C$ , including the equal-width case  $C_{\text{ib}} = C$ . The detail family is assigned to channel subsets of the intermediate feature  $Z \in \mathcal{F}_{C_{\text{ib}}}(\Omega)$ . Let  $\{\Pi^\lambda\}_{\lambda \in \Lambda_{\text{detail}}}$  be pointwise channel-selection operators satisfying*

$$Z = \text{Concat}_{\lambda \in \Lambda_{\text{detail}}} \Pi^\lambda(Z). \quad (57)$$

The channel-partitioned detail response is

$$D_{\Lambda_{\text{detail}}}^{\text{ib}}(Z) = \text{Concat}_{\lambda \in \Lambda_{\text{detail}}} M_{\text{cw}}^\lambda(\Pi^\lambda(Z)). \quad (58)$$

We use the following additive form, which benefits channel efficiency and energy balance:

$$S_\Lambda^{\text{ib}}(Z) = A_{\lambda_0}(Z) + \alpha D_{\Lambda_{\text{detail}}}^{\text{ib}}(Z), \quad \alpha \in \mathbb{R}^{C_{\text{ib}}}. \quad (59)$$

For the transported family  $\Lambda^g$ , we use the induced partition  $\Pi^{\lambda^g} := \Pi^\lambda$ .

This form retains or expands the channel space before spatial mixing, while the channel-partitioned mixer preserves the intermediate width. Its cost is that each detail branch acts only on a subset of channels.

For both forms, we obtain  $\mathfrak{G}_{\text{loc-sim}}$ -covariance at the level of the entire block.

**Proposition 3** ( $\mathfrak{G}_{\text{loc-sim}}$ -Block Covariance). *Let  $\text{Blk}_\Lambda^\star$  be a GeoFormer block from Definition 7, with  $\star \in \{\text{b}, \text{ib}\}$ . Then, for every  $g \in \mathfrak{G}_{\text{loc-sim}}(\Omega)$ , whenever the corresponding mixer covariance is well defined,*

$$\text{Blk}_\Lambda^\star(\rho_{\mathcal{F}_C}(g)F) = \rho_{\mathcal{F}_C}(g)\text{Blk}_{\Lambda^g}^\star(F). \quad (60)$$

*Proof.* By (38), the pointwise channel components  $C_{\text{in}}^\star$  and  $C_{\text{out}}^\star$  commute with the scalar-channel action. Since  $\lambda_0$  has zero angular order, (50) gives covariance of  $A_{\lambda_0}$ , while (52) gives covariance of the detail responses. In the inverted-bottleneck case, the pointwise selectors  $\Pi^\lambda$  and the spatially shared channel scale  $\alpha$  also commute with the action, with  $\Pi^{\lambda^g} := \Pi^\lambda$ , so

$$S_{\Lambda}^{\text{ib}}(\rho_{\mathcal{F}_{\text{Cib}}}(g)Z) = \rho_{\mathcal{F}_{\text{Cib}}}(g)S_{\Lambda^g}^{\text{ib}}(Z). \quad (61)$$

The bottleneck form follows from the same anchor and detail covariances; composing either mixer with the two pointwise channel components proves (60).  $\square$

#### 4.3 $\mathfrak{G}_{\text{loc-sim}}$ -Readout

A stack of  $\mathfrak{G}_{\text{loc-sim}}$ -blocks can learn discriminative spatial features, while these features still retain covariant behavior. In general, such features can be directly used for pixel-level tasks. For image-level tasks, especially classification, however, both the spatial dimension and task-irrelevant covariant behavior should be eliminated. To this end, we define the  $\mathfrak{G}_{\text{loc-sim}}$ -Readout to derive global invariant features.

**Definition 8** ( $\mathfrak{G}_{\text{loc-sim}}$ -Readout). *Let  $F : \mathcal{D} \rightarrow \mathbb{R}^C$  be the terminal feature field before readout. Here  $\mathcal{D}$  is the finite exposed covariant domain, containing all coordinates still acted on at the terminal representation, such as spatial location and, when retained, scale indices. The  $\mathfrak{G}_{\text{loc-sim}}$ -Readout is the global pooling over this exposed covariant domain:*

$$\text{Read}(F) = \frac{1}{|\mathcal{D}|} \sum_{u \in \mathcal{D}} F(u). \quad (62)$$

Note that Read is a commonly used global average pooling in vision architectures when  $\mathcal{D}$  contains only spatial coordinates; therefore, it can be regarded as a generalization on the covariant domain.

**Proposition 4** ( $\mathfrak{G}_{\text{loc-sim}}$ -Readout Invariance). *Let  $g \in \mathfrak{G}_{\text{loc-sim}}(\Omega)$ , and suppose that the induced terminal action on the exposed covariant domain is a bijective reindexing  $\sigma_g : \mathcal{D} \rightarrow \mathcal{D}$ . If  $F$  transforms by pullback on  $\mathcal{D}$ ,*

$$F_g(u) = F(\sigma_g^{-1}u), \quad u \in \mathcal{D}, \quad (63)$$

then

$$\text{Read}(F_g) = \text{Read}(F). \quad (64)$$

*Proof.* Using (63),

$$\begin{aligned} \text{Read}(F_g) &= \frac{1}{|\mathcal{D}|} \sum_{u \in \mathcal{D}} F(\sigma_g^{-1}u) \\ &= \frac{1}{|\mathcal{D}|} \sum_{v \in \mathcal{D}} F(v) = \text{Read}(F), \end{aligned} \quad (65)$$

where the second equality uses that  $\sigma_g$  is a bijection of  $\mathcal{D}$ . This proves (64).  $\square$

Here, the bijection assumption is in fact a natural model of covariant behavior under discrete sampling, but it still involves approximations due to boundary effects and discretization errors. Specifically, we ignore the boundary effects of translations and rotations, as well as the numerical errors caused by inexact grid alignment. For scaling, in addition to these two approximations, there is also a numerical error caused by inexact alignment with the sampled scale grid, which stems from the finite-scale non-closure under scaling. Nevertheless, this is still the most faithful discrete invariance that we can implement.

#### 4.4 GeoFormer Architecture

Up to this point, we have given the complete definitions and properties of the  $\mathfrak{G}_{\text{loc-sim}}$ -Mixer, Block, and Readout, which fully correspond to the theoretical characterization in Sec. 3.

We now turn to the concrete engineering construction of the full GeoFormer architecture. Here, the core learnable component of GeoFormer is our  $\mathfrak{G}_{\text{loc-sim}}$ -block, which accounts for the main learning capacity of the network. Yet, blocks alone are not sufficient to form a typical Transformer.

**Definition 9** (GeoFormer Architecture). *On the input side, blocks operate on embedded tokens produced by Emb. Let  $F_0 = f$ . For stage  $i = 1, \dots, N_{\text{stg}}$ , choose  $\star_i \in \{\text{b}, \text{ib}\}$  and define*

$$\text{Stg}_i = \text{Blk}_{\Lambda_i, N_i}^{\star_i} \circ \dots \circ \text{Blk}_{\Lambda_{i,1}}^{\star_i} \circ \text{Emb}_i. \quad (66)$$

The backbone recursion is

$$F_i = \text{Stg}_i(F_{i-1}), \quad i = 1, \dots, N_{\text{stg}}. \quad (67)$$

On the output side, the tokens produced by deep blocks,  $F_{N_{\text{stg}}}$ , are reduced to a global representation by the  $\mathfrak{G}_{\text{loc-sim}}$ -Readout,

$$\text{Head} = \text{Read}(F_{N_{\text{stg}}}). \quad (68)$$

We also consider a variant of the default GeoFormer, GAFormer, with self-attention stages on  $F_{N_{\text{stg}}}$  to capture global semantics. For  $j = 1, \dots, N_{\text{att}}$ ,

$$\text{AttStg}_j = \text{Att}_{N_j} \circ \dots \circ \text{Att}_1 \circ \text{Emb}_j, \quad (69)$$

where  $\text{Att} = C_{\text{out}} \circ \text{SA} \circ C_{\text{in}}$ , SA is permutation-equivariant self-attention, continuing the recursion as

$$F_{N_{\text{stg}}+j} = \text{AttStg}_j(F_{N_{\text{stg}}+j-1}), \quad (70)$$

with the final  $\mathfrak{G}_{\text{loc-sim}}$ -Readout,

$$\text{Head} = \text{Read}(F_{N_{\text{stg}}+N_{\text{att}}}). \quad (71)$$

In practice, the involved engineering factors such as permutation-equivariant self-attention, residual connections, normalization, activation, LayerScale, and dropout do not introduce additional spatial priors. The spatial prior therefore remains determined by the  $\mathfrak{G}_{\text{loc-sim}}$ -Mixer.

As an overall summary of the theory and architecture, we now position GeoFormer, especially in relation to MetaFormer-like designs:

*GeoFormer can be viewed as a unified architecture for introducing geometric invariance priors into vision Transformers. It extends MetaFormer-like designs to*

TABLE 2  
Implementation Details of GeoFormer in Sec. 5.1

| Item                | Setting                                                   |
|---------------------|-----------------------------------------------------------|
| <i>Architecture</i> |                                                           |
| Blk                 | Bottleneck $\mathcal{G}_{\text{loc-sim}}$ -Block          |
| Emb                 | Shared $1 \times 1$ convolution                           |
| $N_{\text{stg}}$    | 4                                                         |
| $N_{\text{att}}$    | 0                                                         |
| $C_i$               | [32, 96, 288, 864]                                        |
| $N_i$               | [2, 4, 6, 2]                                              |
| $C_{b,i}$           | $\max(16, \lfloor C_i/4 \rfloor)$                         |
| $R_n(r)$            | $\cos(n\pi r)$                                            |
| $\Lambda$           | $\{(n, m, w) \mid n, m \in \{0, 1, 2\}, w \in \{2, 3\}\}$ |
| <i>Training</i>     |                                                           |
| Optimization        | SGD, momentum 0.9, weight decay $10^{-3}$                 |
| Learning rate       | Initial value $10^{-2}$ , $\times 0.1$ every 50 epochs    |
| Training budget     | 100 epochs, batch size 64                                 |
| Loss                | Cross-entropy with label smoothing 0.1                    |

a richer spectrum of invariance priors. This theoretical perspective and principled architecture allow us to achieve concrete trade-offs among invariance, discriminability, and scalability more flexibly in practical tasks, whereas such a theory-to-architecture blueprint is usually lacking in previous studies [26]. Specifically, when the problem scale is small or data are limited, the need for invariance priors may dominate. In this case, one may use equivariant embeddings, while the blocks model the priors required by the task. When the problem scale is large and data are sufficient, invariance priors may instead serve as supplementary design choices. In this case, one may use generic embeddings and global self-attention, while the blocks retain only basic geometric priors.

## 5 EXPERIMENTS

The experiments are organized around two questions that follow directly from the preceding theory and architecture.

**Q1.** Can the invariance–discriminability–scalability trade-off, the pointwise principle, and the spatial-blindness cost established in Sec. 3 be supported by numerical evidence in a controlled experimental setting, thereby demonstrating that the theory is empirically grounded rather than merely formal?

**Q2.** Can the GeoFormer family developed in Sec. 4 achieve effective invariance–discriminability–scalability trade-offs at real-world scale? In particular, can its fixed  $\mathcal{G}_{\text{loc-sim}}$  spatial prior preserve competitive performance and model scalability while providing a clear improvement over heuristic spatial priors, thereby demonstrating that the architecture is practically grounded rather than merely conceptual?

### 5.1 Controlled Evidence for Principles and Costs

This subsection addresses **Q1** through a controlled study on CIFAR-100, providing numerical evidence for the predicted principles and costs under rotation.

#### 5.1.1 Protocol and comparison groups

CIFAR-100 is used as the dataset for this study, with image rotation serving as the basic protocol for both training and testing. Its moderate scale is particularly suitable for

TABLE 3  
CIFAR-100 Accuracy (%) Under Four Training Protocols

| Model                      | N90°        |             | ±180°       |             | 0°          |             | ±15°        |             |
|----------------------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
|                            | Clean       | Robust      | Clean       | Robust      | Clean       | Robust      | Clean       | Robust      |
| <i>Generic backbones</i>   |             |             |             |             |             |             |             |             |
| VGGNet [37]                | 71.4        | 32.3        | 72.2        | 53.4        | 70.8        | 24.3        | 71.3        | 31.3        |
| ResNet [13]                | <b>75.5</b> | 40.7        | <b>76.7</b> | 62.5        | <b>75.5</b> | 29.9        | <b>75.4</b> | 35.4        |
| ViT [3]                    | 51.4        | 30.7        | 53.0        | 39.0        | 54.7        | 23.9        | 53.1        | 26.0        |
| ConvNeXt [29]              | 71.4        | 36.9        | 73.3        | 63.0        | 71.2        | 24.5        | 71.1        | 31.4        |
| PoolFormer [30]            | 67.8        | 39.2        | 68.5        | 54.9        | 69.6        | 29.9        | 70.0        | 33.9        |
| <i>Geometric baselines</i> |             |             |             |             |             |             |             |             |
| WaveNet [35]               | 45.6        | 18.1        | 45.6        | 25.2        | 50.2        | 16.7        | 47.9        | 19.5        |
| GroupNet [4]               | 64.7        | 46.3        | 63.9        | <b>63.3</b> | 58.2        | 43.2        | 61.8        | <b>51.4</b> |
| <i>Invariant baselines</i> |             |             |             |             |             |             |             |             |
| InvPoint [9], [10]         | 36.1        | 34.1        | 37.5        | 35.9        | 39.8        | 35.4        | 38.4        | 34.9        |
| InvConv3 [9], [10]         | 67.5        | 41.5        | 68.0        | 51.7        | 67.6        | 27.5        | 67.2        | 29.5        |
| InvPool [9], [10]          | 55.4        | 48.5        | 54.1        | 51.3        | 55.4        | <b>48.1</b> | 56.1        | 50.1        |
| InvGauss [9], [10]         | 54.3        | <b>51.1</b> | 53.2        | 51.5        | 54.9        | <b>50.3</b> | 54.6        | 51.0        |
| InvShift [9], [10]         | 65.5        | 41.5        | 67.6        | 52.6        | 66.5        | 26.1        | 66.8        | 28.1        |
| <i>Proposed model</i>      |             |             |             |             |             |             |             |             |
| <b>GeoFormer</b>           | <b>72.8</b> | <b>56.9</b> | <b>74.0</b> | <b>68.8</b> | <b>75.0</b> | 46.6        | <b>74.4</b> | <b>54.8</b> |

laboratory-style validation of operator behavior, allowing the relevant trade-offs to be demonstrated under controlled conditions rather than being obscured by excessive data complexity.

**Experimental protocol.** We evaluate each model under four training protocols:

- **N90° protocol:** training uses rotations by multiples of 90° (i.e., 0°, 90°, 180°, 270°);
- **±180° protocol:** training uses arbitrary-angle rotations (i.e.,  $-180^\circ \sim 180^\circ$ );
- **0° protocol:** training uses no rotation augmentation;
- **±15° protocol:** training uses only small-angle rotations (i.e.,  $-15^\circ \sim 15^\circ$ ).

In all four cases, rotation robustness is evaluated on the same arbitrary-angle rotated test set, while clean accuracy is reported on the standard test set. The fixed test-time condition isolates the effect of training-time geometric exposure, allowing us to distinguish architecture-induced robustness from augmentation-dependent robustness. Jointly examining clean and rotated accuracy exposes the discriminability–robustness consequences predicted by the theory.

**Comparison groups.** The compared models are organized according to the role they play in examining **Q1**.

- **Generic backbones:** VGGNet, ResNet, ViT, ConvNeXt, and PoolFormer. These models employ standard learned or heuristic spatial operators. They provide reference points for generic learning and for rotation robustness acquired implicitly from data and augmentation.
- **Geometric baselines:** the wavelet scattering network WaveNet and the group-equivariant network GroupNet. WaveNet represents the front-end/back-end route in Example 1, using a fixed wavelet-scattering front-end followed by a learnable  $1 \times 1$  convolutional back-end. GroupNet represents the group-equivariant route in Sec. 2.3, encoding geometry through equivariant convolutions and prescribed group-structured channels. Both contrast with GeoFormer, which retains scalar channels and distributes geometry through hierarchical spatial mixing.

TABLE 4  
Statistical Indicators of CIFAR-100 Accuracy (%) Under Four Training Protocols

| Model                      | $\mathbb{E}$<br>Clean | $\mathbb{E}$<br>Robust | $\mathbb{E}$<br>C.-R. | $\mathbb{E}$<br>C.&R. |
|----------------------------|-----------------------|------------------------|-----------------------|-----------------------|
| <i>Generic backbones</i>   |                       |                        |                       |                       |
| VGGNet [37]                | 71.4                  | 35.3                   | 36.1                  | 53.4                  |
| ResNet [13]                | <b>75.8</b>           | 42.1                   | 33.6                  | <u>58.9</u>           |
| ViT [3]                    | 53.0                  | 29.9                   | 23.1                  | 41.5                  |
| ConvNeXt [29]              | 71.8                  | 38.9                   | 32.8                  | 55.3                  |
| PoolFormer [30]            | 69.0                  | 39.5                   | 29.5                  | 54.2                  |
| <i>Geometric baselines</i> |                       |                        |                       |                       |
| WaveNet [35]               | 47.3                  | 19.9                   | 27.5                  | 33.6                  |
| GroupNet [4]               | 62.2                  | <u>51.1</u>            | 11.1                  | 56.6                  |
| <i>Invariant baselines</i> |                       |                        |                       |                       |
| InvPoint [9], [10]         | 38.0                  | 35.1                   | <b>2.9</b>            | 36.5                  |
| InvConv3 [9], [10]         | 67.6                  | 37.6                   | 30.0                  | 52.6                  |
| InvPool [9], [10]          | 55.3                  | 49.5                   | 5.7                   | 52.4                  |
| InvGauss [9], [10]         | 54.2                  | 51.0                   | <u>3.3</u>            | 52.6                  |
| InvShift [9], [10]         | 66.6                  | 37.1                   | 29.5                  | 51.8                  |
| <i>Proposed model</i>      |                       |                        |                       |                       |
| <b>GeoFormer</b>           | <u>74.1</u>           | <b>56.8</b>            | 17.3                  | <b>65.4</b>           |

- **Invariant baselines:** these models share the same fixed hierarchical invariant front-end and differ only in the subsequent learning operator or spatial mixer. This controlled setting examines the empirical consequences of Theorem 1 and Corollary 1. We denote them by InvPoint, InvConv3, InvPool, InvGauss, and InvShift. InvPoint uses only shared pointwise channel mixing, realizing the compatible but spatially blind form; InvConv3 introduces unconstrained learnable  $3 \times 3$  spatial interaction; and InvPool, InvGauss, and InvShift retain the same pointwise channel mixing while adding pooling-, Gaussian-, and shift-based spatial components, respectively.
- **GeoFormer:** we adopt the bottleneck realization of the  $\mathcal{G}_{\text{loc-sim}}$ -Block in Example 3. Pointwise embeddings are used to better match the experimental purpose and data scale. The complete architecture and training settings are summarized in Table 2.

### 5.1.2 Main findings

The CIFAR-100 experiments provide controlled numerical evidence for the theoretical characterization in Sec. 3. Across different comparison groups, the results reveal the predicted invariance–discriminability–scalability trade-off: generic learning does not structurally guarantee geometric robustness, pointwise-compatible learning exposes the spatial-blindness cost, and effective spatial interaction is necessary to recover discriminative capacity. Meanwhile, different geometric designs exhibit distinct trade-offs depending on how geometric structure is introduced. Together, these results support the empirical consequences of the pointwise principle, the spatial-blindness cost, and the spatial–channel role division realized by GeoFormer.

Table 3 reports clean and arbitrary-angle rotation accuracy under the four training protocols. Table 4 summarizes the resulting trade-offs, and Fig. 5 visualizes the invariance–discriminability landscape. In Tables 3 and 4, the best and second-best values are highlighted in bold and underlined, respectively.

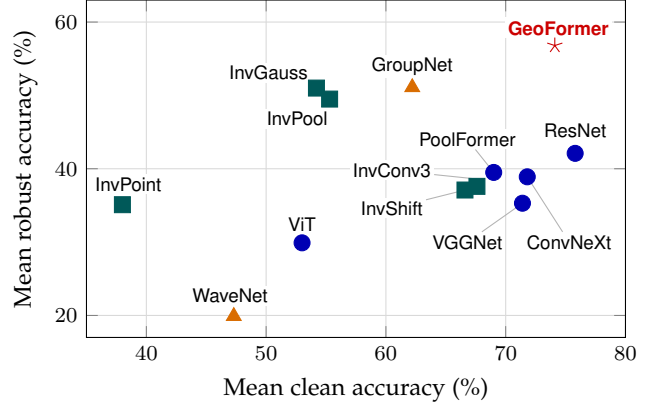


Fig. 5. Visualization of the invariance–discriminability trade-off on CIFAR-100, using mean clean accuracy and mean robust accuracy across the four training protocols. ● generic backbones, ▲ geometric baselines, ■ invariant-front-end baselines, and ★ GeoFormer. GeoFormer lies near the favorable frontier, combining high clean recognition with substantially improved rotational robustness.

### 5.1.3 Generic backbones: implicit invariance

A first pattern is that generic backbones achieve strong clean discrimination but acquire rotation robustness mainly through training-time exposure. ResNet and ConvNeXt are the clearest examples. This dependence on augmentation is consistent with invariance being learned implicitly rather than built into the architecture.

ViT and PoolFormer further clarify the role of spatial priors at this data scale. The weak-prior ViT remains limited, whereas PoolFormer learns more effectively through its lightweight pooling-based spatial prior. However, because this prior does not explicitly encode rotation invariance, PoolFormer still inherits the augmentation dependence characteristic of implicit invariance. Overall, these results reproduce the position of generic backbones in Table 1: they support generic discriminative and scalable learning, but provide limited built-in geometric invariance. This limitation is particularly evident in vision Transformers, where weak spatial and invariance priors make effective learning more difficult at modest data scales.

### 5.1.4 Geometric baselines: explicit invariance under structural constraints

WaveNet and GroupNet represent two established routes for introducing explicit geometry. WaveNet follows the front-end/back-end route, using a fixed invariant front-end followed by a pointwise learnable back-end. Its relatively low clean and rotated accuracies reflect the limited discriminative adaptability of this front-loaded construction.

GroupNet instead encodes geometry through group-equivariant convolutions and prescribed group-structured channels. Its average rotated accuracy of 51.1 confirms the robustness benefit of explicit equivariance, while its clean accuracy remains at 62.2. This follows the regular-representation construction, which allocates one channel per group element so that channel count and memory scale directly with group order [20]. In our implementation, this several-fold memory overhead, together with limited generic channel learning and reduced throughput through

TABLE 5  
Implementation Details of GeoFormer and GAFormer in Sec. 5.2

| Item                | Setting                                                               |
|---------------------|-----------------------------------------------------------------------|
| <i>Architecture</i> |                                                                       |
| Blk                 | Inverted-bottleneck $\mathfrak{G}_{\text{loc-sim}}$ -Block            |
| Emb                 | Strided convolutional embedding                                       |
| $N_{\text{stg}}$    | 4 (GeoFormer) / 2 (GAFormer)                                          |
| $N_{\text{att}}$    | 0 (GeoFormer) / 2 (GAFormer)                                          |
| $C_{\text{ib}}$     | $C_i$ (GeoFormer) / $2C_i$ (GAFormer)                                 |
| $R_n(r)$            | $\cos(n\pi r)$                                                        |
| $\Lambda$           | $\{(n, m, w) \mid n, m \in \{0, 1, 2\}, w = 2\}$                      |
| <i>Training</i>     |                                                                       |
| Input resolution    | $224 \times 224$                                                      |
| Training budget     | 300 epochs, global batch size 2048                                    |
| Optimization        | AdamW, weight decay 0.05                                              |
| Learning rate       | Peak $1.5 \times 10^{-3}$ , 5-epoch warmup, cosine decay to $10^{-6}$ |
| Data augmentation   | RandAugment, Mixup 0.8, CutMix 1.0, random erasing 0.25               |
| Regularization      | Label smoothing 0.1 and stochastic depth                              |

specialized convolutions, makes ImageNet-scale training difficult and reflects the partial scalability in Table 1.

### 5.1.5 Invariant baselines: the principle and cost

The invariant baselines provide the most controlled evidence for the pointwise principle and spatial-blindness cost. They share the same fixed hierarchical invariant front-end and differ only in the subsequent learning operator or spatial component, thereby isolating the effects of pointwise and spatial learning.

InvPoint realizes the shared pointwise form selected by Theorem 1. Its clean and rotated accuracies remain similarly low, at 38.0 and 35.1 on average, respectively. This is consistent with Corollary 1: pointwise channel mixing preserves the existing geometric structure but cannot renew discriminative spatial information.

InvConv3 introduces unconstrained learnable  $3 \times 3$  spatial interaction. Its average clean accuracy rises to 67.6, but the clean-robust gap increases to 30.0, showing that spatial freedom restores discrimination while weakening geometric control.

InvPool, InvGauss, and InvShift further show that spatial interaction is necessary, while its form determines the resulting trade-off. Pooling and Gaussian mixing preserve stronger rotation robustness but remain limited in clean discrimination, whereas shift mixing recovers more clean accuracy with weaker robustness. Together, these results reveal the predicted spectrum from spatially blind pointwise learning to increasingly expressive but less geometry-controlled spatial interaction.

### 5.1.6 GeoFormer: the role division for a favorable trade-off

The CIFAR-100 results thus provide controlled numerical evidence that GeoFormer can achieve a concrete, task-dependent position in the trade-off summarized in Table 1. For this setting, GeoFormer is configured as an invariance-prior-dominant realization: bottleneck  $\mathfrak{G}_{\text{loc-sim}}$ -blocks, pointwise embeddings, and an invariant readout, while retaining generic channel learning. The resulting model achieves average clean and rotated accuracies of 74.1 and 56.8, respectively, giving the highest joint score in Table 4. It also obtains the best rotated accuracy under  $\pm 180^\circ$ ,  $N90^\circ$ , and  $\pm 15^\circ$ . Under  $0^\circ$ , several invariant baselines

TABLE 6  
Scale-Specific GeoFormer and GAFormer Configurations in Sec. 5.2

| Scale                                                                     | $C_i, C_j$          | $N_i, N_j$     | Params (M) | MACs (G)   |
|---------------------------------------------------------------------------|---------------------|----------------|------------|------------|
| <i>GeoFormer (Geo, Geo, Geo, Geo); <math>\Delta</math> vs. PoolFormer</i> |                     |                |            |            |
| S12                                                                       | [64, 128, 320, 512] | [2, 2, 6, 2]   | 11.9(+0.0) | 1.9(+0.1)  |
| S24                                                                       | [64, 128, 320, 512] | [4, 4, 12, 4]  | 21.4(+0.0) | 3.5(+0.1)  |
| S36                                                                       | [64, 128, 320, 512] | [6, 6, 18, 6]  | 30.8(+0.0) | 5.2(+0.2)  |
| M36                                                                       | [96, 192, 384, 768] | [6, 6, 18, 6]  | 56.1(+0.0) | 9.1(+0.3)  |
| M48                                                                       | [96, 192, 384, 768] | [8, 8, 24, 8]  | 73.4(+0.0) | 12.0(+0.4) |
| <i>GAFormer (Geo, Geo, Att, Att); <math>\Delta</math> vs. PAFormer</i>    |                     |                |            |            |
| S18                                                                       | [64, 128, 320, 512] | [3, 3, 9, 3]   | 26.3(+0.0) | 4.1(+0.1)  |
| S36                                                                       | [64, 128, 320, 512] | [3, 12, 18, 3] | 39.1(+0.0) | 8.0(+0.2)  |
| M36                                                                       | [96, 192, 384, 576] | [3, 12, 18, 3] | 56.0(+0.0) | 13.2(+0.2) |

achieve higher rotated accuracy only at a substantial loss of clean recognition. This result illustrates the practical role of the spatial-channel division derived in Sec. 3.3, thereby supporting Q1.

## 5.2 Large-Scale Evidence for the GeoFormer

This subsection addresses Q2 through a large-scale study on ImageNet-1K, providing numerical evidence for whether the GeoFormer family can introduce geometric invariance priors while preserving the discriminative capacity and scalability required by modern vision backbones.

### 5.2.1 Protocol and comparison groups

ImageNet-1K is used as the dataset for this study, with real-world clean and robustness evaluation under standard augmentation training. Its scale and diversity are particularly suitable for testing the scalability of GeoFormer, even with a fixed  $\mathfrak{G}_{\text{loc-sim}}$  prior.

**Experimental protocol.** We train all internal models on the standard ImageNet-1K training set using the same strong augmentation and optimization recipe. Results for external models are taken from the cited sources and may use different training recipes. Performance on the standard validation set is used primarily to assess the overall competitiveness of each architecture and its behavior under model scaling. The comparisons cover a broad range of model scales and are conducted under fair or approximately matched architectural and computational settings.

We further evaluate the trained models directly, without any additional fine-tuning or adaptation, on ImageNet-A [38], a widely used and challenging robustness benchmark composed of naturally occurring hard examples selected to expose failures of standard ImageNet classifiers, thereby reflecting real-world distribution shifts rather than any single artificial degradation. Under architecturally aligned comparisons in which the spatial mixer is the principal variable, this evaluation is used primarily to assess the practical robustness and generalization associated with different spatial priors.

**Comparison groups.** The compared models are organized into three groups according to their roles in addressing Q2.

- **External positioning models:** we organize the external baselines into two broad groups. The first uses the classical ResNet architecture, instantiated as RSB-ResNet, which retains a purely convolutional backbone while adopting a modern Transformer-era training recipe. The second covers representative backbones

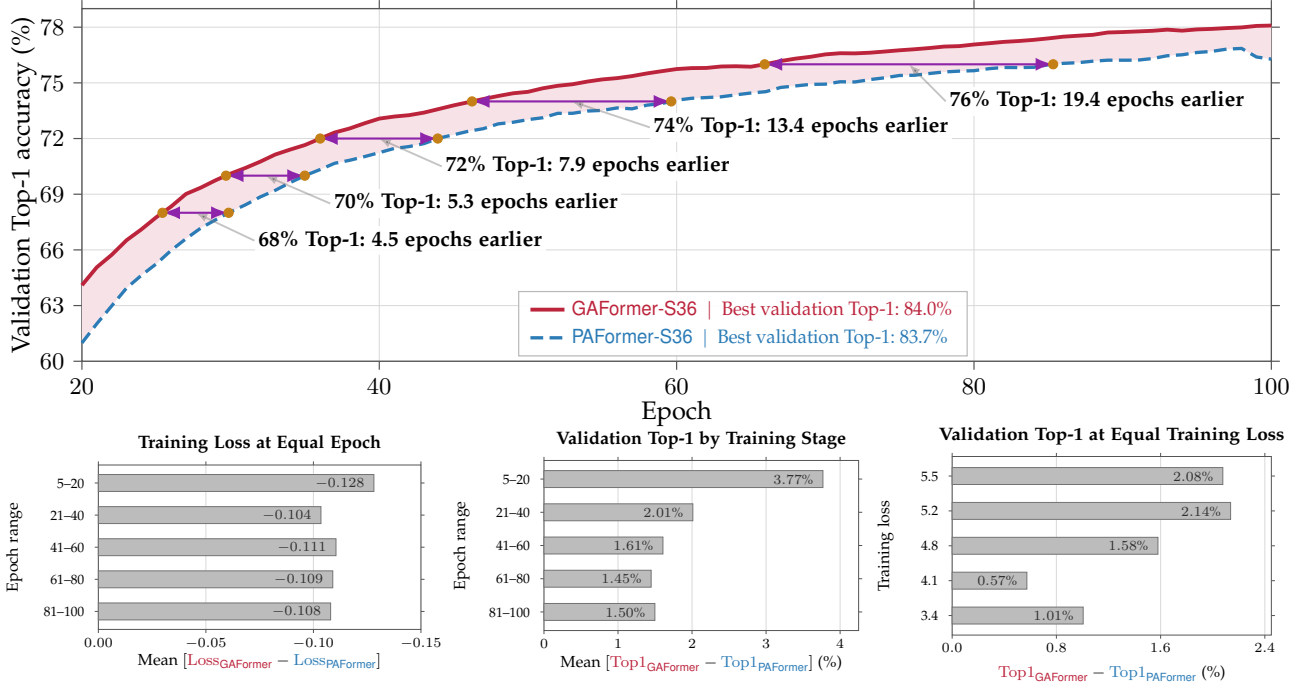


Fig. 6. Visualization of the training dynamics and matched statistics of GAFormer and PAFormer on ImageNet. During early training (epochs 20–100), GAFormer exhibits greater training sample efficiency and lower generalization loss: it reaches specified accuracy milestones substantially earlier and consistently outperforms PAFormer in all three statistics, namely training loss at matched epochs, validation accuracy at matched epochs, and validation accuracy at matched training loss. The two architectures are tightly aligned, isolating the  $\mathcal{G}_{\text{loc-sim}}$ -Mixer as the source of these measured efficiency and generalization gains.

developed in the Transformer era, spanning attention-based Transformers (ViT, DeiT, PVT, Swin, RVT, FAN, CSWin, and MViTv2), convolution–attention hybrids (UniFormer and MaxViT), and modern convolutional designs shaped by the same evolving design principles (RegNetY, ConvNeXt, and VAN). Together, these models contextualize ImageNet performance across representative architectural paradigms and model scales.

- **Internal MetaFormer models:** we subsume the MetaFormer family into the geometric-prior spectrum of GeoFormer and evaluate the robustness and scalability of a series of spatial mixers under tightly aligned architectural settings. IdentityFormer contains no spatial mixer, whereas RandFormer introduces global random mixing into the last two stages of IdentityFormer; both therefore serve as baselines without a structured spatial prior. PoolFormer employs a pooling-based spatial mixer and can be viewed as a special case of GeoFormer corresponding to the zeroth-order form of the  $\mathcal{G}_{\text{loc-sim}}$ -Mixer; it therefore serves as an important heuristic spatial-prior baseline. ConvFormer instead uses learnable convolutional spatial mixers, representing a data-adaptive spatial prior distilled from real-world-scale data. For PoolFormer, ConvFormer, and GeoFormer, global self-attention can further be introduced into the last two stages for high-level semantic abstraction, yielding PAFormer, CAFormer, and GAFormer, respectively. These attention-hybrid variants allow us to examine the discriminative ceiling attainable under each given low-level spatial prior.
- **GeoFormer and GAFormer:** We adopt the inverted-bottleneck realization of the  $\mathcal{G}_{\text{loc-sim}}$ -Block in Exam-

ple 4, together with standard convolutional embeddings, to accommodate the experimental objective and data scale. GAFormer additionally incorporates self-attention stages; their detailed configurations are provided in Tables 5 and 6. As discussed above, the  $\mathcal{G}_{\text{loc-sim}}$  prior is only one possible realization rather than a defining restriction of the framework. In this sense, GeoFormer generalizes MetaFormer along the dimension of spatial priors. Here, we examine whether GeoFormer can retain favorable model scalability even when equipped with a fixed and highly specialized spatial mixer. This provides a real-world-scale test of whether the principles and costs derived by our theory remain practically relevant. By contrast, comparable geometric priors are difficult to scale within the WaveNet and GroupNet frameworks, neither of which performs effectively on ImageNet-1K. Our emphasis is therefore not on whether GeoFormer surpasses the state of the art—for that purpose, a weaker and more data-adaptive spatial prior would be more appropriate, as discussed above. Instead, we ask whether the theoretically derived prior improves upon heuristic spatial priors and how closely it approaches the data-adaptive spatial prior learned from real-world-scale data. These comparisons provide empirical support for the spatial-prior spectrum described in Example 2.

### 5.2.2 Main findings

The ImageNet experiments provide large-scale numerical evidence for the architectural implementation in Sec. 4.

- **Efficiency.** The  $\mathcal{G}_{\text{loc-sim}}$  prior improves training sample efficiency and generalization, enabling faster acquisi-

TABLE 7  
Parameter Counts, MACs, and ImageNet Clean Top-1 Accuracy of  
Convolutional-Style Backbones

| Model                     | Params (M) | MACs (G) | Clean Acc. (%)                |
|---------------------------|------------|----------|-------------------------------|
| <i>S12 scale</i>          |            |          |                               |
| RSB-ResNet-18 [13], [39]  | 11.7       | 1.8      | 70.6                          |
| IdentityFormer-S12 [36]   | 11.9       | 1.8      | 74.6                          |
| RandFormer-S12 [36]       | 12.1       | 1.9      | 76.6                          |
| PoolFormer-S12 [30]       | 11.9       | 1.8      | <u>77.2</u>                   |
| <b>GeoFormer-S12</b>      | 11.9       | 1.9      | <b>78.4</b> <sub>(+1.2)</sub> |
| <i>S24 scale</i>          |            |          |                               |
| RSB-ResNet-34 [13], [39]  | 21.8       | 3.7      | 75.5                          |
| IdentityFormer-S24 [36]   | 21.3       | 3.4      | 78.2                          |
| RandFormer-S24 [36]       | 21.8       | 3.5      | 78.8                          |
| PoolFormer-S24 [30]       | 21.4       | 3.4      | <u>80.3</u>                   |
| <b>GeoFormer-S24</b>      | 21.4       | 3.5      | <b>80.6</b> <sub>(+0.3)</sub> |
| <i>S36 scale</i>          |            |          |                               |
| RSB-ResNet-50 [13], [39]  | 25.6       | 4.1      | 79.8                          |
| IdentityFormer-S36 [36]   | 30.8       | 5.0      | 79.3                          |
| RandFormer-S36 [36]       | 31.5       | 5.2      | 79.5                          |
| PoolFormer-S36 [30]       | 30.8       | 5.0      | <u>81.4</u>                   |
| <b>GeoFormer-S36</b>      | 30.8       | 5.2      | <b>81.6</b> <sub>(+0.2)</sub> |
| <i>M36 scale</i>          |            |          |                               |
| RSB-ResNet-101 [13], [39] | 44.5       | 7.9      | 81.3                          |
| IdentityFormer-M36 [36]   | 56.1       | 8.8      | 80.0                          |
| RandFormer-M36 [36]       | 56.8       | 9.0      | 81.2                          |
| PoolFormer-M36 [30]       | 56.1       | 8.8      | <u>82.1</u>                   |
| <b>GeoFormer-M36</b>      | 56.1       | 9.1      | <b>82.2</b> <sub>(+0.1)</sub> |
| <i>M48 scale</i>          |            |          |                               |
| RSB-ResNet-152 [13], [39] | 60.2       | 11.6     | 81.8                          |
| IdentityFormer-M48 [36]   | 73.3       | 11.5     | 80.4                          |
| RandFormer-M48 [36]       | 74.2       | 11.9     | 81.4                          |
| PoolFormer-M48 [30]       | 73.4       | 11.6     | <u>82.5</u>                   |
| <b>GeoFormer-M48</b>      | 73.4       | 12.0     | <b>82.7</b> <sub>(+0.2)</sub> |

tion of discriminative representations with fewer sample exposures.

- **Discriminability.** Despite using a fixed  $\mathcal{G}_{\text{loc-sim}}$  mixer, GeoFormer and GAFormer achieve strong ImageNet clean accuracy and approach the empirical performance ceiling represented by highly data-adaptive backbones such as CAFormer.
- **Robustness.** The  $\mathcal{G}_{\text{loc-sim}}$  prior yields strong ImageNet robust accuracy, consistently outperforming heuristic spatial priors in aligned comparisons and remaining competitive with modern fully learned backbones.
- **Scalability.** These advantages persist across increasing model scales, while earlier explicit-geometric routes such as WaveNet and GroupNet encounter discriminative or computational bottlenecks at ImageNet scale.

Together, these results support the practical effectiveness of GeoFormer in realizing favorable invariance–discriminability–scalability trade-offs at real-world scale.

In the following tables, the best and second-best results at each scale are highlighted in bold and underlined, respectively, and the subscripts of GeoFormer and GAFormer report their gains over the tightly aligned PoolFormer and PAFormer counterparts.

### 5.2.3 Efficiency

Fig. 6 provides process-level evidence that is not visible from the final ImageNet accuracy alone. Under the tightly aligned GAFormer-S36 and PAFormer-S36 architectures, GAFormer reaches the same validation-accuracy milestones substantially earlier, with the lead becoming larger toward higher milestones. The matched statistics show the

TABLE 8  
Parameter Counts, MACs, and ImageNet Clean Top-1 Accuracy of  
Modern Vision Backbones in the Transformer Era

| Model                  | Params (M) | MACs (G) | Clean Acc. (%)                |
|------------------------|------------|----------|-------------------------------|
| <i>S18 scale</i>       |            |          |                               |
| DeiT-S [22]            | 22.0       | 4.6      | 79.8                          |
| PVT-Small [40]         | 25.0       | 3.8      | 79.8                          |
| RegNetY-4G [39], [41]  | 21.0       | 4.0      | 81.3                          |
| VAN-B2 [42]            | 27.0       | 5.0      | 82.8                          |
| Swin-T [27]            | 29.0       | 4.5      | 81.3                          |
| RVT-S [43]             | 23.0       | 4.7      | 81.9                          |
| ConvNeXt-T [29]        | 29.0       | 4.5      | 82.1                          |
| FAN-S [44]             | 28.0       | 5.3      | 82.5                          |
| CSWin-T [45]           | 23.0       | 4.3      | 82.7                          |
| MViTv2-T [46]          | 24.0       | 4.7      | 82.3                          |
| ConvFormer-S18 [36]    | 27.0       | 3.9      | 83.0                          |
| CAFormer-S18 [36]      | 26.3       | 4.1      | <b>83.6</b>                   |
| PAFormer-S18 [36]      | 26.3       | 4.0      | 82.8                          |
| <b>GAFormer-S18</b>    | 26.3       | 4.1      | <u>83.2</u> <sub>(+0.4)</sub> |
| <i>S36 scale</i>       |            |          |                               |
| PVT-Medium [40]        | 44.0       | 6.7      | 81.2                          |
| RegNetY-8G [39], [41]  | 39.0       | 8.0      | 82.1                          |
| VAN-B3 [42]            | 45.0       | 9.0      | 83.9                          |
| Swin-S [27]            | 50.0       | 8.7      | 83.0                          |
| ConvNeXt-S [29]        | 50.0       | 8.7      | 83.1                          |
| FAN-B [44]             | 54.0       | 10.4     | 83.6                          |
| CSWin-S [45]           | 35.0       | 6.9      | 83.6                          |
| MViTv2-S [46]          | 35.0       | 7.0      | 83.6                          |
| UniFormer-B [47]       | 50.0       | 8.3      | 83.9                          |
| ConvFormer-S36 [36]    | 40.0       | 7.6      | <u>84.1</u>                   |
| CAFormer-S36 [36]      | 39.3       | 8.0      | <b>84.5</b>                   |
| PAFormer-S36 [36]      | 39.1       | 7.8      | 83.7                          |
| <b>GAFormer-S36</b>    | 39.1       | 8.0      | 84.0 <sub>(+0.3)</sub>        |
| <i>M36 scale</i>       |            |          |                               |
| ViT-B/16 [3]           | 86.0       | 17.6     | 79.7                          |
| PVT-Large [40]         | 61.0       | 9.8      | 81.7                          |
| RegNetY-16G [39], [41] | 84.0       | 15.9     | 82.2                          |
| VAN-B4 [42]            | 60.0       | 12.2     | 84.2                          |
| Swin-B [27]            | 88.0       | 15.4     | 83.5                          |
| RVT-B [43]             | 92.0       | 17.7     | 82.6                          |
| ConvNeXt-B [29]        | 89.0       | 15.4     | 83.8                          |
| FAN-L [44]             | 81.0       | 15.8     | 83.9                          |
| CSWin-B [45]           | 78.0       | 15.0     | 84.2                          |
| MViTv2-B [46]          | 52.0       | 10.2     | 84.4                          |
| MaxViT-S [48]          | 69.0       | 11.7     | <u>84.5</u>                   |
| ConvFormer-M36 [36]    | 57.0       | 12.8     | <b>84.5</b>                   |
| CAFormer-M36 [36]      | 56.2       | 13.2     | <b>85.2</b>                   |
| PAFormer-M36 [36]      | 56.0       | 13.0     | 84.2                          |
| <b>GAFormer-M36</b>    | 56.0       | 13.2     | <u>84.5</u> <sub>(+0.3)</sub> |

same pattern throughout early training. At equal epochs, GAFormer maintains lower training loss and higher validation accuracy. At equal training loss, it still achieves higher validation accuracy. These results show that the  $\mathcal{G}_{\text{loc-sim}}$ -Mixer improves both training sample efficiency and generalization.

This process-level advantage is important for interpreting the final discriminability results. With sufficient training and sample exposure, generic learning gradually narrows the final accuracy gap between GAFormer and PAFormer. Fig. 6 shows that the advantage of GAFormer is much more pronounced under a smaller training budget. This observation directly reflects the statistical motivation of invariance in the Introduction: an informative geometric prior reduces the sample exposure required to acquire discriminative representations.

### 5.2.4 Discriminability

Tables 7 and 8 examine whether explicit geometric priors can reach the discriminative regime of modern ImageNet backbones. The models listed in these tables are mostly

TABLE 9  
Parameter Counts, MACs, and ImageNet Robust Top-1 Accuracy of  
Convolutional-Style Backbones

| Model                     | Params (M) | MACs (G) | Robust Acc. (%)               |
|---------------------------|------------|----------|-------------------------------|
| <i>S12 scale</i>          |            |          |                               |
| RSB-ResNet-18 [13], [39]  | 11.7       | 1.8      | 2.0                           |
| IdentityFormer-S12 [36]   | 11.9       | 1.8      | 5.4                           |
| RandFormer-S12 [36]       | 12.1       | 1.9      | <u>7.1</u>                    |
| PoolFormer-S12 [30]       | 11.9       | 1.8      | 6.5                           |
| <b>GeoFormer-S12</b>      | 11.9       | 1.9      | <b>9.0</b> <sub>(+2.5)</sub>  |
| <i>S24 scale</i>          |            |          |                               |
| RSB-ResNet-34 [13], [39]  | 21.8       | 3.7      | 4.4                           |
| IdentityFormer-S24 [36]   | 21.3       | 3.4      | 9.7                           |
| RandFormer-S24 [36]       | 21.8       | 3.5      | 13.2                          |
| PoolFormer-S24 [30]       | 21.4       | 3.4      | <u>14.0</u>                   |
| <b>GeoFormer-S24</b>      | 21.4       | 3.5      | <b>15.5</b> <sub>(+1.5)</sub> |
| <i>S36 scale</i>          |            |          |                               |
| RSB-ResNet-50 [13], [39]  | 25.6       | 4.1      | 7.5                           |
| IdentityFormer-S36 [36]   | 30.8       | 5.0      | 13.4                          |
| RandFormer-S36 [36]       | 31.5       | 5.2      | 16.9                          |
| PoolFormer-S36 [30]       | 30.8       | 5.0      | <u>18.1</u>                   |
| <b>GeoFormer-S36</b>      | 30.8       | 5.2      | <b>20.6</b> <sub>(+2.5)</sub> |
| <i>M36 scale</i>          |            |          |                               |
| RSB-ResNet-101 [13], [39] | 44.5       | 7.9      | 13.9                          |
| IdentityFormer-M36 [36]   | 56.1       | 8.8      | 17.4                          |
| RandFormer-M36 [36]       | 56.8       | 9.0      | 20.4                          |
| PoolFormer-M36 [30]       | 56.1       | 8.8      | <u>22.2</u>                   |
| <b>GeoFormer-M36</b>      | 56.1       | 9.1      | <b>26.0</b> <sub>(+3.8)</sub> |
| <i>M48 scale</i>          |            |          |                               |
| RSB-ResNet-152 [13], [39] | 60.2       | 11.6     | 17.3                          |
| IdentityFormer-M48 [36]   | 73.3       | 11.5     | 19.1                          |
| RandFormer-M48 [36]       | 74.2       | 11.9     | 21.9                          |
| PoolFormer-M48 [30]       | 73.4       | 11.6     | <u>23.6</u>                   |
| <b>GeoFormer-M48</b>      | 73.4       | 12.0     | <b>27.9</b> <sub>(+4.3)</sub> |

strong modern baselines, whereas the historical baselines for explicit geometry occupy a very different performance regime. WaveNet and GroupNet remain meaningful geometric baselines on CIFAR-100, but their limitations become pronounced on ImageNet-1K. The ImageNet performance reported for the WaveNet route remains below 60%. GroupNet instead relies on prescribed group-structured channels and specialized equivariant convolutions, with several-fold memory overhead and very low throughput. These two routes therefore fail to reach the discriminative baseline region of modern ImageNet backbones. GeoFormer moves explicit geometry into this region and reaches a highly competitive position.

Table 7 provides the cleaner structural comparison. Within the tightly aligned MetaFormer scaffold, IdentityFormer, RandFormer, PoolFormer, and GeoFormer represent progressively different spatial priors: no structured prior, random mixing, a heuristic prior, and the highly designed  $\mathfrak{G}_{\text{loc-sim}}$  prior. GeoFormer achieves the best clean accuracy at every reported scale and consistently improves upon PoolFormer. This ordering forms an empirical performance ladder along the spatial-prior spectrum.

Table 8 further tests whether this advantage remains after self-attention is introduced. GAFormer stays in the leading performance group across all reported scales – ranking second behind CAFormer at S18 and M36, and only marginally below second at S36 – a notable position given the strong Transformer-era competition with highly learned or data-adaptive spatial processing. CAFormer functions here as the empirical ceiling: unified timm evaluations independently place the CAFormer family near the top of

TABLE 10  
Parameter Counts, MACs, and ImageNet Robust Top-1 Accuracy of  
Modern Vision Backbones in the Transformer Era

| Model               | Params (M) | MACs (G) | Robust Acc. (%)               |
|---------------------|------------|----------|-------------------------------|
| <i>S18 scale</i>    |            |          |                               |
| DeiT-S [22], [49]   | 22.0       | 4.6      | 18.7                          |
| Swin-T [27]         | 29.0       | 4.5      | 21.6                          |
| RVT-S [43]          | 23.0       | 4.7      | 25.7                          |
| ConvNeXt-T [29]     | 29.0       | 4.5      | 24.2                          |
| FAN-S [44]          | 28.0       | 5.3      | 29.1                          |
| ConvFormer-S18 [36] | 27.0       | 3.9      | 25.3                          |
| CAFormer-S18 [36]   | 26.3       | 4.1      | <b>33.5</b>                   |
| PAFormer-S18 [36]   | 26.3       | 4.0      | 29.4                          |
| <b>GAFormer-S18</b> | 26.3       | 4.1      | <u>29.5</u> <sub>(+0.1)</sub> |
| <i>S36 scale</i>    |            |          |                               |
| Swin-S [27]         | 50.0       | 8.7      | 32.3                          |
| ConvNeXt-S [29]     | 50.0       | 8.7      | 31.2                          |
| FAN-B [44]          | 54.0       | 10.4     | 35.4                          |
| ConvFormer-S36 [36] | 40.0       | 7.6      | 33.2                          |
| CAFormer-S36 [36]   | 39.3       | 8.0      | <b>40.9</b>                   |
| PAFormer-S36 [36]   | 39.1       | 7.8      | 34.0                          |
| <b>GAFormer-S36</b> | 39.1       | 8.0      | <u>36.0</u> <sub>(+2.0)</sub> |
| <i>M36 scale</i>    |            |          |                               |
| Swin-B [27]         | 88.0       | 15.4     | 35.8                          |
| RVT-B [43]          | 92.0       | 17.7     | 28.5                          |
| ConvNeXt-B [29]     | 89.0       | 15.4     | 36.7                          |
| FAN-L [44]          | 81.0       | 15.8     | 37.2                          |
| MaxViT-S [48], [49] | 69.0       | 11.7     | 37.7                          |
| ConvFormer-M36 [36] | 57.0       | 12.8     | 37.6                          |
| CAFormer-M36 [36]   | 56.2       | 13.2     | <b>45.6</b>                   |
| PAFormer-M36 [36]   | 56.0       | 13.0     | 37.2                          |
| <b>GAFormer-M36</b> | 56.0       | 13.2     | <u>40.0</u> <sub>(+2.8)</sub> |

the ImageNet-1K regime [49], and it is the only model that consistently remains above GAFormer across all three scales. The geometric advantage identified in the attention-free comparison thus persists in modern architectures, with the fixed  $\mathfrak{G}_{\text{loc-sim}}$  prior approaching this data-adaptive frontier.

Together, the two tables reveal a broader spatial-prior spectrum: no prior, random prior, heuristic prior, the highly designed geometric prior instantiated by GeoFormer/GAFormer, and the data-adaptive prior represented by ConvFormer/CAFormer. The resulting performance hierarchy provides large-scale empirical support for the interpretation developed in Sec. 3.3.

### 5.2.5 Robustness

Tables 9 and 10 provide the corresponding ImageNet robustness evidence, following the same attention-free/attention-hybrid split as above.

In the attention-free group, GeoFormer achieves the best robust accuracy at every reported scale, improving on the tightly aligned PoolFormer by 2.5, 1.5, 2.5, 3.8, and 4.3 points from S12 to M48 – several times the 0.1–1.2-point clean-accuracy gains shown in Table 7. In the attention-hybrid group, GAFormer ranks second in robust accuracy at every scale – 0.1, 2.0, and 2.8 points above PAFormer at S18, S36, and M36 – consistently surpassed only by CAFormer.

Together with the discriminability results, this indicates that the  $\mathfrak{G}_{\text{loc-sim}}$  prior’s contribution is weighted toward robustness rather than clean accuracy: GAFormer stays close to the clean-performance frontier while holding the second overall robustness position among the strong modern backbones considered here.

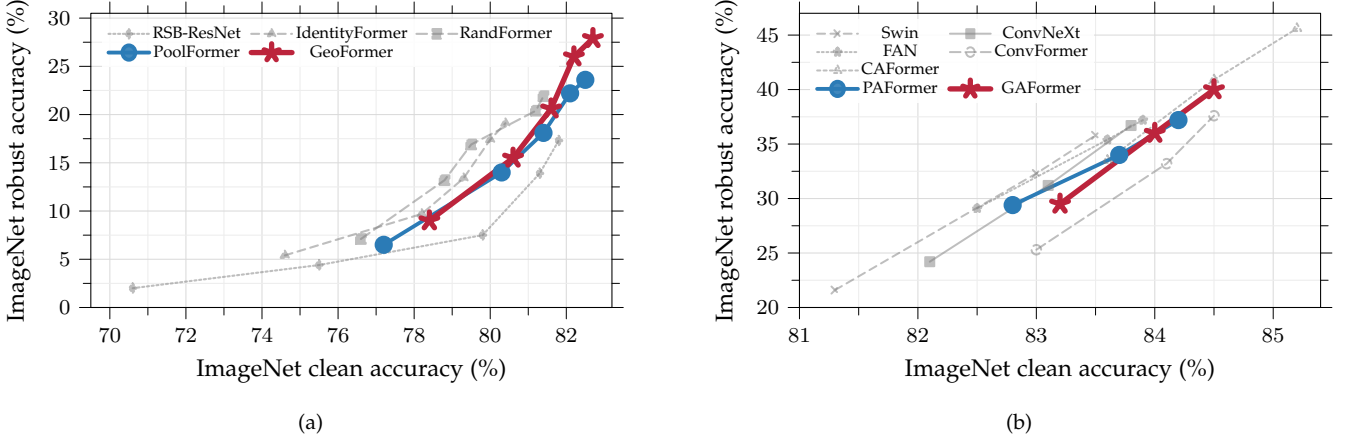


Fig. 7. ImageNet clean–robust accuracy trajectories under model scaling, jointly visualizing invariance, discriminability, and scalability. (a) In the tightly controlled setting, the  $\Phi_{\text{loc-sim}}$ -Mixer achieves a more favorable three-way trade-off than heuristic mixers. (b) In modern backbones, its invariance advantage over heuristic mixing persists while its scaling trajectory remains within a bounded gap of fully data-adaptive mixers. Together, the two views support the spatial-prior spectrum in Example 2.

### 5.2.6 Scalability

Tables 7–10 already report clean and robust endpoints at multiple model scales. Fig. 7 connects these discrete results into scaling trajectories and shows how the operating position associated with each spatial prior evolves with model scale.

Panel (a) provides the cleaner attention-free view. All model families move toward higher clean and robust accuracy as scale increases, while GeoFormer remains consistently above PoolFormer along the trajectory. The clean advantage becomes smaller at larger scales, whereas the robustness separation remains clear and becomes more pronounced toward the larger models. This behavior is consistent with the efficiency results above. Greater model capacity and longer training progressively narrow part of the final discriminative difference, while the structural contribution of the geometric prior to robust recognition remains visible. The geometric advantage therefore persists as a scaling behavior across model sizes.

Panel (b) extends the same view to modern attention-hybrid backbones. The GAFormer trajectory remains consistently more favorable than that of the aligned PAFormer and enters the leading clean–robust region occupied by highly data-adaptive models. Other strong backbones, including Swin, ConvNeXt, FAN, and ConvFormer, either remain lower in robustness or do not provide a better joint clean–robust operating point. The resulting GAFormer trajectory therefore lies second only to the highly data-adaptive CAFormer in the joint clean–robust comparison.

These trajectories provide the direct large-scale evidence for the invariance–discriminability–scalability trade-off posed in Q2.

### 5.2.7 Ablation

Table 11 checks whether the preceding behavior depends on a specific realization of the  $\Phi_{\text{loc-sim}}$ -Mixer. Multi-scale and higher-order variants preserve or improve the clean–robust performance of the basic realization at both the S12 and M48 scales. The main findings therefore persist across different mixer realizations, supporting the geometric spatial prior as the relevant structural factor.

TABLE 11  
Ablations of the  $\Phi_{\text{loc-sim}}$  Mixer with Multi-scale and Higher-order Variants

| Variant                           | Clean Acc. (%) | Robust Acc. (%) |
|-----------------------------------|----------------|-----------------|
| <i>Small scale: GeoFormer-S12</i> |                |                 |
| Basic                             | 78.4           | 9.0             |
| Multi-scale variant               | 79.1           | 9.8             |
| Higher-order variant              | 79.4           | 11.9            |
| <i>Large scale: GeoFormer-M48</i> |                |                 |
| Basic                             | 82.7           | 27.9            |
| Multi-scale variant               | 82.9           | 29.6            |
| Higher-order variant              | 82.8           | 28.0            |

## 6 CONCLUSION

The four historical routes to geometric invariance in vision – invariants, implicit CNNs, equivariant CNNs, and now Transformers – are usually told as mutually competing paradigms (Table 1). This paper argues instead that they are different settlements of one underlying trade-off among invariance, discriminability, and scalable learning. We give that trade-off, for the first time, a principled form and a structural cost. Once the trade-off has this explicit form, how much of it to pay stops being a matter of taste.

The principled form is pointwise. On the scalar channels already standard in vision Transformers,  $\Phi_{\text{loc-sim}}$  compatibility forces any regular local operator to collapse into shared pointwise channel mixing (Theorem 1). This reinterprets the ubiquitous  $1 \times 1$  convolution: it is a geometric necessity, not just an engineering default. The structural cost follows from the same collapse. Such an operator can never renew local spatial discrimination, so a hierarchy built from it alone stays permanently blind to its own neighborhood (Corollary 1). This is why the MetaFormer-like separation between spatial and channel mixing is a structural trade-off.

Guided by the principles and costs, we proposed GeoFormer as a unified architecture for introducing geometric invariance priors into vision Transformers. It extends MetaFormer-like designs to a richer spectrum of invariance priors (Example 2), specifically bridging the gap between heuristic priors and full data-adaptive convolution priors

with  $\mathfrak{G}_{\text{loc-sim}}$ -Mixer, Block, and Readout. GeoFormer allows us to achieve concrete trade-offs among invariance, discriminability, and scalability more flexibly in practical tasks, supported by substantial numerical evidence on CIFAR-100 (Fig. 5) and ImageNet (Fig. 7).

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